

Handbook of Triaxial Hopkinson Pressure Bar Testing and Experiments

*Theory, Apparatus, and the Virtual Tri-HB
Integrated Computational Workspace*

*A reference for postgraduate students, researchers,
and laboratory practitioners*

Qianbing Zhang

Department of Civil and Environmental Engineering
Monash University
Wellington Road, Clayton, VIC 3800, Australia
qianbing.zhang@monash.edu

First Edition

May 29, 2026

Handbook of Triaxial Hopkinson Pressure Bar Testing and Experiments
First edition.

© 2026 Qianbing Zhang. All rights reserved.

This handbook is the companion reference for the `tri_hb_integrated.py` computational workspace developed within the Tri-HB Research Group, Monash University. Reproduction of any part of this work for commercial purposes requires the written permission of the author. Reproduction for academic teaching and non-commercial research is permitted provided appropriate citation is given.

Suggested citation:

Zhang, Q. (2026). *Handbook of Triaxial Hopkinson Pressure Bar Testing and Experiments*. First Edition. Department of Civil and Environmental Engineering, Monash University, Clayton, VIC, Australia.

Disclaimer. The procedures described herein involve the acceleration of solid bodies to high velocities and the release of large amounts of stored electrical and elastic energy. The author and Monash University accept no liability for personal injury, equipment damage, or research outcomes arising from the use of this material. Readers are responsible for ensuring that any physical implementation conforms to the safety regulations of their host institution.

Preface

This handbook documents the theory, apparatus, and computational workflow of the triaxial Hopkinson pressure bar (Tri-HB) facility developed by the Rock Mechanics and Dynamics group at Monash University, together with its digital twin, the *Virtual Tri-HB Integrated Workspace* (`tri_hb_integrated.py`). The material is intended as a self-contained reference for postgraduate students in geomechanics, mining engineering, civil engineering, and applied mechanics, and as a working document for laboratory practitioners commissioning or using comparable equipment.

The Tri-HB facility extends the classical split Hopkinson pressure bar (SHPB) of Kolsky [1] along three orthogonal axes, enabling controlled dynamic loading on cubic specimens under independent triaxial pre-stress. Four loading modes are documented: (i) the classical gas-gun mode, in which a projectile launches a compressive pulse on a single bar; (ii) electromagnetic uniaxial mode, in which the projectile is replaced by a programmable electromagnetic pulse generator that produces a clean half-sine waveform of independently tunable amplitude and duration; (iii) electromagnetic asynchronous triaxial mode, in which three orthogonal pulses fire with independent time delays, allowing the influence of loading order on dynamic failure to be isolated; and (iv) electromagnetic symmetric multidirectional mode, in which opposing bar pairs fire simultaneously and the resulting constructive wave superposition delivers specimen stresses that are inaccessible in any one-sided configuration without bar plasticity.

The handbook is organised in three parts. Part I (Chapters 1 and 2) develops the theoretical foundations common to all modes: the one-dimensional wave equation in elastic bars, force and velocity continuity at specimen faces, the three-wave method, and the algebra of acoustic impedance. Part II (Chapters 3–9) treats each of the six loading modes in turn, deriving the relevant wave equations from first principles and presenting worked numerical examples. Part III (Chapters 10–12) documents the integrated computational workspace, its data-reduction routines, and the wave–stress–path–damage validation pipeline.

Every equation in the handbook is derived from declared first principles. A deliberate choice has been made to retain the elementary derivations even where more abstract or compact forms exist, on the grounds that the handbook is intended to be *used* for laboratory work and student supervision. Each worked example has been independently checked against the deployed digital twin, and the parameter ranges quoted (bar elastic limit, pre-stress envelope, strain-rate window) reflect the calibration of the Monash facility at the time of publication.

The author thanks members of the Monash Rock Mechanics and Dynamics group for discussions during the development of the Tri-HB facility and the computational workspace, and the postgraduate students whose questions over several teaching cycles informed the explanatory structure adopted here. Corrections, suggestions, and reports of typographical errors are gratefully received at qianbing.zhang@monash.edu.

Qianbing Zhang
Clayton, Victoria
May 29, 2026

Contents

Preface	iii
Nomenclature	ix
1 Introduction	1
1.1 The Monash Tri-HB system	2
1.2 Scope and assumed background	2
1.3 Notation conventions	3
2 Theoretical foundations	4
2.1 The one-dimensional wave equation in the bar	4
2.1.1 Acoustic impedance	4
2.1.2 Reflection and transmission at a planar interface	5
2.1.3 Pochhammer–Chree dispersion	5
2.2 Force and velocity continuity at a specimen face	5
2.3 The three-wave method	6
2.3.1 The one-wave (transmitted-wave) form	6
2.3.2 Stress equilibrium criterion	6
2.3.3 Specimen aspect ratio	7
2.4 Bar plasticity envelope	7
2.5 Triaxial pre-stress on the three orthogonal axes	7
2.6 Dynamic increase factor	8
2.7 Specimen lateral response and Poisson coupling	8
2.8 Elastic moduli used in the analysis	9
2.9 Bar-wave energy partitioning	9
3 Mode 1: Gas-gun uniaxial impact (classical SHPB)	10
3.1 Hardware configuration	10
3.2 Generation of the incident pulse	11
3.2.1 The pulse shaper	11
3.3 Pulse propagation and specimen response	11
3.4 Reading the strain gauge records	12
3.5 Wave analysis: the three-wave method applied	12
3.6 Lateral response on Y and Z	12
3.7 Validity envelope	12

4	Mode 2: Confinement-chamber SHPB	13
4.1	Hardware configuration	14
4.2	Loading sequence	14
4.3	Wave analysis on the active (X) axis	15
4.4	Stress invariants and Lode angle	15
4.5	Confinement strengthening	15
4.6	Two parasitic corrections	15
4.6.1	Chamber compliance	15
4.6.2	Jacket parasitic load	16
4.7	Procedure	16
4.8	Worked example: sandstone in a steel chamber	17
4.9	Validity envelope	17
5	Mode 3: Monash Tri-HB — gas-gun coupled static–dynamic triaxial loading	19
5.1	Hardware configuration	19
5.2	Loading sequence and superposition	20
5.2.1	Why the bar gauges do not record the pre-stress	20
5.3	Wave analysis on the active (X) axis	21
5.4	Confinement strengthening	21
5.5	Procedure	21
5.6	Worked example: sandstone under triaxial pre-stress	22
5.7	Validity envelope	23
6	Mode 4: Electromagnetic uniaxial impact	24
6.1	Hardware differences from Mode 1	24
6.2	The half-sine pulse	25
6.2.1	Frequency content of the half-sine	25
6.3	Step-by-step procedure	25
6.4	Worked example: sandstone with triaxial pre-stress	26
7	Mode 5: Electromagnetic asynchronous triaxial impact	27
7.1	Hardware configuration	27
7.2	The time-delay concept	28
7.3	Three-wave analysis on each axis	28
7.4	Procedure	28
7.5	Validity notes	29

8	Mode 6: Electromagnetic symmetric multidirectional impact	30
8.1	Symmetry of the wave field	31
8.2	Derivation of the symmetric-mode wave equations	31
8.3	Comparison: standard vs symmetric mode	32
8.4	Symmetric sub-modes	33
8.5	Pressure–volume response under hydrostatic loading	33
8.6	Procedure	33
8.7	Worked example: concrete under fully hydrostatic loading	34
9	Comparison of the six loading modes	35
9.1	Side-by-side comparison	35
9.2	Mode selection: matching the question to the apparatus	35
9.3	Common pitfalls and digital-twin diagnostics	36
10	The integrated computational workspace	38
10.1	Launching the application	38
10.2	Top-level navigation	39
10.3	Step 1 sidebar: the two tabs	39
10.4	Main panel tabs in the Step 1 simulator	40
10.5	Bar-plasticity warning banner	40
10.6	Data export from Step 1	41
10.7	Worked Streamlit session	41
11	Experimental data analysis and reduction	43
11.1	Three data sources	43
11.2	Equations used by the reducer	43
11.3	Reducer outputs and energy sanity check	45
11.4	Worked reducer session	45
12	Wave model, stress path, damage and DEM validation	46
12.1	Wave-model equations (Step 2 tab)	46
12.1.1	Elastic constants and wave speeds	46
12.1.2	Normalised delay and regime classifier	46
12.1.3	Superposition factor at the specimen centre	47
12.2	Stress-path equations (Step 3 tab)	47
12.2.1	Failure surface and failure index	48
12.2.2	Equivalent (deviatoric) strain rate	48
12.3	Damage and energy equations (Step 4 tab)	48

12.3.1	Damage evolution	49
12.3.2	Stiffness and wave-speed degradation	49
12.3.3	Energy indicators per unit volume	49
12.3.4	Synthetic DEM/experimental descriptors	50
12.4	Sidebar controls common to Steps 2–4	50
12.5	Export from Step 4	51
A	Reference tables	52
A.1	Bar properties (42CrMo steel) – default values	52
A.2	Default specimen parameters in the simulator	52
A.3	Default damage-model parameters (Step 4 Advanced mode)	53
A.4	Useful identities	53
B	Suggested exercises	54
	References	56

Nomenclature

Roman symbols

Symbol	Meaning	Units
A	Failure-surface intercept constant	MPa
A_b	Bar cross-sectional area	m ²
A_s	Specimen cross-sectional area	m ²
a_θ	Lode-angle amplitude in failure surface	–
B	Failure-surface pressure coefficient	–
b_{rate}	Rate-hardening coefficient (DIF slope)	–
c_p	Constrained (P-wave) wave speed in the specimen	m/s
c_s	Shear (S-wave) wave speed in the specimen	m/s
C_0	Longitudinal bar wave speed, $C_0 = \sqrt{E_b/\rho_b}$	m/s
D	Scalar damage variable	–
$\dot{D}$	Damage rate	s ⁻¹
D_c	Central damage fraction (DEM descriptor)	–
d	Bar or specimen diameter	m
E, E_b, E_s	Young’s modulus (general, bar, specimen)	Pa
F	Failure index, $F = q/q_f$	–
F_0	Failure-index normalisation	–
F_1, F_2	Force at the two specimen faces	N
f, g	d’Alembert wave functions (right- and left-going)	–
G	Shear modulus	Pa
$h(\theta)$	Lode-angle correction in failure surface	–
J_2, J_3	Second and third invariants of the deviatoric stress	MPa ² , MPa ³
K	Bulk modulus	Pa
k_{conf}	Mohr–Coulomb confinement coefficient	–
L_g	Strain-gauge distance from specimen face	m
L_s	Specimen loading length	m
L_{striker}	Striker bar length	m
M	Constrained (P-wave) modulus	Pa
m	Overstress exponent in damage law	–
n	Failure-surface pressure exponent	–
p	Mean (hydrostatic) pressure, $p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$	MPa
q	Mises equivalent (deviatoric) stress	MPa
q_f	Failure-surface deviatoric stress at given p, θ	MPa
r	Reflection / superposition amplitude ratio	–
S_x	Damage symmetry index (DEM descriptor)	–
s_i	Deviatoric stress component, $s_i = \sigma_i - p$	MPa
t	Time	s
t_{eq}	Stress-equilibrium time window	s
t_{travel}	Wave travel time across specimen, $t_{\text{travel}} = L_s/c_p$	s
$u(x, t)$	Axial displacement in the bar	m
V	Striker impact velocity	m/s

Symbol	Meaning	Units
V_s	Specimen volume, $V_s = A_s L_s$	m^3
v	Particle velocity at specimen face	m/s
W	Energy (per unit volume or total)	J or J/m^3
x	Axial coordinate along bar	m
Z	Acoustic impedance, $Z = \rho c$	$\text{kg m}^{-2} \text{s}^{-1}$

Greek symbols

Symbol	Meaning	Units
α	Damage saturation exponent	—
β	Damage rate exponent	—
Δt	Inter-axis pulse delay	s
Δt^*	Normalised delay, $\Delta t^* = \Delta t / t_{\text{travel}}$	—
ε	Strain (axial, specimen, or bar)	—
$\varepsilon_I, \varepsilon_R, \varepsilon_T$	Incident, reflected, transmitted bar strains	—
$\varepsilon_I^\pm$	Incident strain in the $\pm$ direction (symmetric mode)	—
ε_v	Volumetric strain	—
$\dot{\varepsilon}$	Strain rate	s^{-1}
$\dot{\varepsilon}_{\text{eq}}$	Equivalent (deviatoric) strain rate	s^{-1}
$\dot{\varepsilon}_0$	Reference strain rate in damage law	s^{-1}
η_{sup}	Superposition factor at specimen centre	—
θ	Lode angle	rad
ν	Poisson's ratio	—
ρ, ρ_b, ρ_s	Density (general, bar, specimen)	kg/m^3
σ	Stress (axial, specimen, or bar)	MPa
$\sigma_1, \sigma_2, \sigma_3$	Principal pre-stresses on the three orthogonal axes	MPa
σ_{c0}	Quasi-static uniaxial compressive strength	MPa
σ_{peak}	Peak amplitude of incident pulse	MPa
σ_{prop}	Bar proportional limit	MPa
σ_Y	Bar yield stress	MPa
τ	Pulse duration	s
τ_D	Damage time scale	s

Abbreviations

Abbreviation	Meaning
DEM	Discrete Element Method
DIF	Dynamic Increase Factor
EM	Electromagnetic
SHPB	Split Hopkinson Pressure Bar
Tri-HB	Triaxial Hopkinson Bar
UCS	Uniaxial Compressive Strength

Introduction

The split Hopkinson pressure bar (SHPB), conceived by Kolsky in 1949 [1] on the foundation laid by John and Bertram Hopkinson, has become the standard apparatus for measuring the stress-strain response of solids in the strain-rate window $\dot{\epsilon} \in 10^2 \text{ s}^{-1}$ to 10^4 s^{-1} [2, 3]. For geomaterials and brittle solids, where the dynamic strength can be several times the quasi-static value, the SHPB is the principal source of constitutive data informing rock blasting, tunnel stability under seismic load, projectile impact, and rockburst hazard [4, 5].

The conventional SHPB is a one-dimensional instrument: a striker bar launches a compressive pulse along a single axis. Real engineering loadings, however, are rarely uniaxial. Blast loading drives multiaxial pulses with finite phase lag; in-situ rockmasses sit under static triaxial confinement before they receive the dynamic loading; and the constitutive behaviour of porous rocks under simultaneous deviatoric and volumetric loading cannot be inferred from uniaxial data alone. These limitations motivated the development of triaxial Hopkinson bar (Tri-HB) facilities [6, 7], in which three pairs of orthogonal bars apply independently controlled dynamic pulses on a cubic specimen held under independently controlled triaxial pre-stress.

The Tri-HB facility documented in this handbook supports six loading modes:

1. **Mode 1 – Gas-gun uniaxial** (classical SHPB). A pneumatically launched striker bar strikes the incident bar, producing a roughly trapezoidal pulse shaped by a copper or rubber pulse shaper. The specimen is loaded along a single axis, with no active confining pressure.
2. **Mode 2 – Confinement-chamber SHPB**. A cylindrical specimen is held inside a pressurised confining chamber (oil-filled vessel or elastomeric sleeve) that delivers axisymmetric lateral pressure $\sigma_2 = \sigma_3 = p_c$. The axial baseline σ_1 is applied through a hydraulic cylinder on the X line, as in the Monash system, while the dynamic pulse is launched by the gas gun on the incident bar. This is the most widely used triaxial-SHPB configuration in the literature, but it cannot probe non-axisymmetric stress states.
3. **Mode 3 – Monash Tri-HB**: gas-gun coupled static-dynamic triaxial loading. Same gas-gun X-axis hardware as Modes 1 and 2, but replaces the passive confining chamber with active orthogonal bar pairs and hydraulic cylinders on the Y and Z axes. This delivers independently controlled $\sigma_1, \sigma_2, \sigma_3$ and live monitoring of the lateral wave response on four output bars [6, 7], enabling true triaxial dynamic loading of cubic specimens.
4. **Mode 4 – Electromagnetic uniaxial**. A capacitor-bank-driven coil generates a clean half-sine pulse on a single bar, with independently tunable amplitude and duration.
5. **Mode 5 – Electromagnetic asynchronous triaxial**. Three orthogonal pulses fire with programmable inter-axis delays, exposing the path-dependence of dynamic failure.
6. **Mode 6 – Electromagnetic symmetric multidirectional**. Opposing bar pairs fire simultaneously, constructively superposing in force while cancelling in momentum and doubling the achievable specimen stress without exceeding the bar elastic limit [8].

All six modes permit independent static pre-stress on the three principal axes (axisymmetric in Mode 2; fully independent in Modes 3–6), in the range 0 MPa to 50 MPa per axis for the current Monash digital twin.

The handbook is intended to be read in conjunction with the *Virtual Tri-HB Integrated Workspace*, a Streamlit-based digital twin (file `tri_hb_integrated.py`) that bundles four analysis steps into one shared experimental setup: (1) configuration and simulation or reduction of bar-gauge data; (2) wave propagation modelling and equilibrium analysis; (3) stress-path and invariant analysis; and (4) damage evolution and DEM/ experimental validation. The workspace is described in detail in Chapters 10–12.

1.1 The Monash Tri-HB system

The experimental work documented in this handbook was carried out on the triaxial Hopkinson bar system at Monash University, used for dynamic uniaxial, biaxial, and triaxial strength testing of rocks [6, 7]. The system comprises a dynamic loading subsystem (a gas gun and a cylindrical striker bar), three independent pairs of square steel bars (42CrMo steel, cross-section 50 mm×50 mm) aligned orthogonally along the X, Y, and Z directions, three hydraulic cylinders (pressure capacity up to 50 MPa), a strong platform, six high-strength steel reaction frames, and a multi-channel high-speed data-acquisition system.

Along the X (axial) direction, the gas-gun loading train consists of a striker barrel (1.5 m), an incident bar (2.5 m), a transmission bar (2 m), an absorption bar (0.5 m), a hydraulic load cylinder, and a moment-trap device. Along the Y and Z directions, four steel output bars (2 m each) apply confining pressure through hydraulic load cylinders and simultaneously monitor the output waves. Overall, the apparatus occupies 8 m in the horizontal X direction, 5 m in the horizontal Y direction, and 5 m in the vertical Z direction.

During a test, quasi-static pre-stresses are applied independently in the three orthogonal directions on a cubic specimen by two horizontal hydraulic cylinders and one vertical hydraulic cylinder, while the dynamic load is delivered by launching the striker bar from the gas gun. This arrangement realises triaxial static–dynamic coupled loading, allowing the influence of the triaxial quasi-static pre-stress on the dynamic behaviour of rocks to be investigated systematically. The six loading modes described above are realised on this physical platform (Modes 1–3, with the cylindrical specimen of Mode 2 and the cubic specimen of Mode 3) and on its electromagnetic extension and digital twin (Modes 4–6).

1.2 Scope and assumed background

The handbook assumes familiarity with linear elasticity, continuum mechanics at the level of an undergraduate solid mechanics course, and elementary numerical methods. Specialised topics — dispersion correction, damage mechanics, stress-path analysis, and discrete element validation — are developed from first principles where required.

The principal omissions, which the reader should consult elsewhere, are: high-frequency Pochhammer–Chree dispersion at the wavelength level [9, 10], the design of pulse shapers [11], and the detailed electromagnetic design of the pulse generator. These are treated in the references cited.

1.3 Notation conventions

Stresses are reported in MPa unless otherwise stated. Compressive stresses and strains are taken as *positive*, in keeping with the geomechanics literature. For bar wave analysis, the incident wave ε_I is therefore positive, and the reflected wave ε_R at a softer specimen is *negative* (tensile). The transmitted wave ε_T is positive for a compressive specimen response. Where a result is more naturally expressed in the tensile-positive convention of elasticity, the convention is stated explicitly at the point of use.

Time is in seconds, except in worked examples where microseconds are used for clarity. Strain rates are in s^{-1} .

Chapter 2

Theoretical foundations

This chapter develops the equations common to all six loading modes. The treatment follows the classical exposition of Kolsky [1, 12] with notational alignment to the modern SHPB literature [13, 14].

2.1 The one-dimensional wave equation in the bar

Each pressure bar in the Tri-HB system is a long, slender rod of high-strength steel (typically 42CrMo). Provided the dominant wavelength is much greater than the bar diameter — a condition satisfied for all pulses considered here — the axial displacement $u(x, t)$ obeys the one-dimensional elastic wave equation [15]:

$$\frac{\partial^2 u}{\partial t^2} = C_0^2 \frac{\partial^2 u}{\partial x^2}, \quad C_0 = \sqrt{\frac{E_b}{\rho_b}}. \quad (2.1)$$

For 42CrMo steel ($E_b = 210$ GPa, $\rho_b = 7850$ kg/m³) the nominal bar wave speed is $C_0 \approx 5172$ m/s. The Step 1 sidebar of the digital twin exposes E_b and C_0 as editable inputs so that a different bar material, or a dispersion-corrected calibrated speed, may be entered without code changes.

The general solution of (2.1) is the d'Alembert decomposition

$$u(x, t) = f(x - C_0 t) + g(x + C_0 t), \quad (2.2)$$

where f propagates in the $+x$ direction and g in the $-x$ direction. The corresponding axial strain and particle velocity are

$$\varepsilon(x, t) = \frac{\partial u}{\partial x} = f'(x - C_0 t) + g'(x + C_0 t), \quad (2.3)$$

$$v(x, t) = \frac{\partial u}{\partial t} = -C_0 f'(x - C_0 t) + C_0 g'(x + C_0 t). \quad (2.4)$$

For a wave travelling purely in the $+x$ direction these collapse to the *Hopkinson identity*

$$\boxed{v(x, t) = -C_0 \varepsilon(x, t)}, \quad (2.5)$$

which is the cornerstone of bar-gauge analysis: the particle velocity is directly proportional to the strain measured by the gauge, with the bar wave speed as the constant of proportionality.

2.1.1 Acoustic impedance

The acoustic impedance of a linear elastic medium is

$$Z = \rho c, \quad (2.6)$$

where ρ is the density and c the relevant wave speed. For a 42CrMo steel bar $Z_b \approx 40.6 \times 10^6$ kg m⁻² s⁻¹. The product $E_b A_b / C_0 = Z_b A_b$ has units of mass-flux per unit time, often called the *mechanical impedance* of the bar; it sets the scale of force/velocity coupling at every interface.

2.1.2 Reflection and transmission at a planar interface

When a plane wave in medium 1 (impedance Z_1) impinges normally on the interface with medium 2 (impedance Z_2), continuity of traction and velocity yields the classical Fresnel-type relations for stress amplitudes [15]:

$$R = \frac{\sigma_R}{\sigma_I} = \frac{Z_2 - Z_1}{Z_2 + Z_1}, \quad T = \frac{\sigma_T}{\sigma_I} = \frac{2Z_2}{Z_1 + Z_2}. \quad (2.7)$$

For a soft specimen ($Z_s < Z_b$) between two steel bars, $R < 0$: the reflected pulse is tensile, which is the source of the negative-going ε_R recorded in compression SHPB tests on brittle solids. For a stiff specimen ($Z_s > Z_b$), $R > 0$: the reflected pulse is compressive. This sign-flip is a diagnostic check on the analysis.

2.1.3 Pochhammer–Chree dispersion

The one-dimensional wave equation (2.1) assumes that all axial fibres of the bar move in phase, which holds only in the long-wavelength limit. The exact solution for an axisymmetric infinite cylinder of radius a is the Pochhammer–Chree mode of dispersion [9, 10], in which the phase velocity c_ϕ depends on the dimensionless wavenumber ka :

$$c_\phi(ka) = C_0 \Phi(ka, \nu_b), \quad (2.8)$$

where Φ is computed from the lowest root of the Pochhammer–Chree characteristic equation, and ν_b is the bar Poisson’s ratio. For $ka \lesssim 0.1$, $\Phi \approx 1$ to within a fraction of one percent; for $ka \sim 1$, Φ decreases by typically 5%, depending on ν_b . The simulator’s default of an unshaped rectangular pulse contains arbitrarily high ka harmonics and is therefore prone to spurious oscillations as the pulse propagates; the half-sine pulse of the electromagnetic mode and the trapezoidal shape of the pulse-shaped gas-gun mode both concentrate energy at $ka \ll 1$ and are essentially dispersion-free for the geometries used here.

A working rule is that bar dispersion is negligible provided

$$\frac{a}{c_\phi \tau_r} \lesssim 0.1, \quad (2.9)$$

where τ_r is the pulse rise time. For a 50 mm square bar ($a_{\text{eq}} \approx 28$ mm) with $C_0 = 5172$ m/s, a rise time $\tau_r > 50$ μ s satisfies this rule comfortably.

2.2 Force and velocity continuity at a specimen face

Each face of the specimen is bonded acoustically to one bar end. At the incident-bar / specimen interface, the wave field in the bar is

$$\varepsilon_{\text{bar}}(x, t) = \varepsilon_I(t - x/C_0) + \varepsilon_R(t + x/C_0), \quad (2.10)$$

where the second term represents the wave reflected from the specimen back into the bar. Force and velocity at the bar end (taken as $x = 0$) are

$$F_1(t) = E_b A_b [\varepsilon_I(t) + \varepsilon_R(t)], \quad (2.11)$$

$$v_1(t) = C_0 [\varepsilon_I(t) - \varepsilon_R(t)]. \quad (2.12)$$

On the opposite specimen face, the transmission bar carries only an outgoing (rightward-propagating) wave ε_T :

$$F_2(t) = E_b A_b \varepsilon_T(t), \quad (2.13)$$

$$v_2(t) = C_0 \varepsilon_T(t). \quad (2.14)$$

Sign convention

$\varepsilon_I > 0$: compressive incident wave. $\varepsilon_R < 0$: tensile reflection (specimen softer than bar, the usual case). $\varepsilon_T > 0$: compressive transmission. For a stiff specimen ($Z_s > Z_b$), $\varepsilon_R > 0$ instead, and this serves as a diagnostic of impedance match.

2.3 The three-wave method

The specimen, of cross-section A_s and loading length L_s , is small compared with the bar wavelength. After a few internal reverberations ($t \gtrsim 3 t_{\text{travel}}$ with $t_{\text{travel}} = L_s/c_p$ the wave travel time across the specimen), dynamic stress equilibrium $F_1 \approx F_2$ is achieved. Under this condition, the specimen stress and strain rate follow from the two-face boundary conditions (2.11)–(2.14):

$$\sigma_s(t) = \frac{F_1 + F_2}{2A_s} = \frac{E_b A_b}{2A_s} [\varepsilon_I(t) + \varepsilon_R(t) + \varepsilon_T(t)], \quad (2.15)$$

$$\dot{\varepsilon}_s(t) = \frac{v_1 - v_2}{L_s} = \frac{C_0}{L_s} [\varepsilon_I(t) - \varepsilon_R(t) - \varepsilon_T(t)], \quad (2.16)$$

$$\varepsilon_s(t) = \int_0^t \dot{\varepsilon}_s(\tau) d\tau. \quad (2.17)$$

These are the *three-wave equations*. They are universal: they hold whether the pulse is launched by a striker bar, an electromagnetic generator, or any other source, and they require no constitutive assumption about the specimen. The bars serve as diagnostic instruments only.

2.3.1 The one-wave (transmitted-wave) form

Under perfect stress equilibrium $\varepsilon_I + \varepsilon_R = \varepsilon_T$, the three-wave equations collapse to:

$$\sigma_s(t) = \frac{E_b A_b}{A_s} \varepsilon_T(t), \quad \dot{\varepsilon}_s(t) = -\frac{2C_0}{L_s} \varepsilon_R(t). \quad (2.18)$$

The stress comes from a single bar gauge (the transmitted-wave gauge); the strain rate, from the reflected gauge alone. Equations (2.15) and (2.18) should agree in the equilibrium window; their divergence is a quantitative measure of non-equilibrium [16]. The Step 1 data reducer of the digital twin offers both forms.

2.3.2 Stress equilibrium criterion

The standard quantitative criterion for stress equilibrium [16, 17] is

$$\mathcal{R}(t) \equiv \frac{|F_1(t) - F_2(t)|}{\frac{1}{2}(F_1(t) + F_2(t))} < 0.05, \quad (2.19)$$

that is, the force imbalance is less than 5% of the mean force. The time at which this is first achieved scales with the wave travel time [18]:

$$t_{\text{eq}} \approx (3 \text{ to } 5) t_{\text{travel}} = (3 \text{ to } 5) \frac{L_s}{c_p}. \quad (2.20)$$

For granite ($c_p \approx 4700 \text{ m/s}$) and $L_s = 50 \text{ mm}$, $t_{\text{travel}} \approx 10.6 \mu\text{s}$, giving $t_{\text{eq}} \approx 32 \mu\text{s}$ to $53 \mu\text{s}$. Brittle specimens that fail before t_{eq} should be tested with a slower rise time or a shorter L_s .

2.3.3 Specimen aspect ratio

For uniaxial compression tests, friction at the bar/specimen interfaces introduces lateral constraint that artificially inflates the apparent strength. The standard recommendation [18, 13] is a specimen aspect ratio

$$\frac{L_s}{d_s} \in [0.5, 1.0], \quad (2.21)$$

with the lower bound favoured for high-rate tests (to minimise t_{travel}) and the upper bound for low-rate or quasi-static comparisons. Lubrication of the specimen ends (PTFE shim, MoS₂ grease) further reduces the friction correction. The 50 mm cubic specimens used in the Tri-HB facility correspond to $L_s/d_s = 1.0$ on each axis.

2.4 Bar plasticity envelope

The three-wave method, and indeed all of the analysis in this handbook, assumes that the bars remain linearly elastic. In the current Streamlit digital twin, the 42CrMo bar proportional limit is set to $\sigma_{\text{prop}} = 1000$ MPa. Whenever the stress carried by any bar exceeds this value, the wave equation (2.1) is no longer valid, and the gauge signal cannot be interpreted by linear analysis. This single constraint defines the operating envelope of the facility:

$$\boxed{\sigma_{\text{bar}}^{\text{max}} \leq \sigma_{\text{prop}} = 1000 \text{ MPa.}} \quad (2.22)$$

The digital twin implements a three-tier visual warning system: *caution* at $0.85 \sigma_{\text{prop}}$ (yellow), *warning* at σ_{prop} (orange), and *critical* at the yield stress $\sigma_Y = 1080$ MPa (red). Every test, simulated or planned, should clear the caution tier.

2.5 Triaxial pre-stress on the three orthogonal axes

Every Tri-HB test admits independent quasi-static pre-stress on each of the three principal axes, applied through servo-controlled hydraulic cylinders *before* the dynamic pulse arrives. The three pre-stresses are denoted σ_1 on the X (loading) axis, σ_2 on Y, and σ_3 on Z, with slider range 0 MPa to 50 MPa per axis in the current digital-twin calibration.

The static and dynamic stress contributions superpose linearly during the dynamic event:

$$\boxed{\begin{aligned} \sigma_X^{\text{total}}(t) &= \sigma_1^{\text{static}} + \sigma_X^{\text{dyn}}(t), \\ \sigma_Y^{\text{total}}(t) &= \sigma_2^{\text{static}} + \sigma_Y^{\text{dyn}}(t), \\ \sigma_Z^{\text{total}}(t) &= \sigma_3^{\text{static}} + \sigma_Z^{\text{dyn}}(t). \end{aligned}} \quad (2.23)$$

The roles played by the three pre-stresses are physically distinct:

- **σ_1 is an axial baseline.** Because X is the loading direction, σ_1 does not strengthen the specimen against the dynamic pulse: it shifts the entire σ - ϵ curve upward by a constant. Increasing σ_1 raises the recorded total axial stress but does not change the peak *rock* strength.
- **σ_2 and σ_3 are confining stresses.** Because Y and Z are perpendicular to X, they confine the specimen and *strengthen* it through the Mohr–Coulomb mechanism. To a good approximation, for low-to-moderate confinement,

$$\sigma_{\text{peak}}^{\text{rock}} = (\sigma_{c0} + k_{\text{conf}} \max(\sigma_2, \sigma_3)) \cdot \text{DIF}, \quad (2.24)$$

where DIF is the dynamic-increase factor introduced in §2.6.

Why the bar gauges do not record the pre-stress

The static pre-stress is held by the hydraulic cylinders and reacted by the bars and the specimen as a single static elastic system at zero frequency. The bar gauges are calibrated to the pre-event baseline, which is *subtracted* during data acquisition. The wave-analysis equations operate exclusively on the dynamic component; the static pre-stress is added back at the reporting stage through (2.23).

2.6 Dynamic increase factor

The strength of geomaterials at high strain rates exceeds the quasi-static value. For uniaxial compression, the rate dependence is conventionally described by a Dynamic Increase Factor (DIF) of the form [19, 20]

$$\text{DIF} = \frac{\sigma_c(\dot{\varepsilon})}{\sigma_c(\dot{\varepsilon}_{\text{ref}})} = 1 + b_{\text{rate}} \log_{10} \left(\frac{\max(\dot{\varepsilon}, \dot{\varepsilon}_{\text{ref}})}{\dot{\varepsilon}_{\text{ref}}} \right), \quad \dot{\varepsilon}_{\text{ref}} = 1 \times 10^{-5} \text{ s}^{-1}. \quad (2.25)$$

The current Streamlit implementation uses this single linear-log DIF law across the simulator's planning strain-rate window. Bilinear or sigmoidal high-rate DIF laws can be used in a separate calibration study, but they are not applied by the default simulator presets. Representative rate-hardening coefficients b_{rate} and reference rates $\dot{\varepsilon}_{\text{ref}}$ are tabulated in Appendix A.

2.7 Specimen lateral response and Poisson coupling

The Monash Tri-HB and the electromagnetic triaxial modes (Modes 3–6) carry *two* output bars on each lateral axis — one bonded to each opposing face of the cubic specimen. The lateral expansion of the specimen drives an outgoing compressive wave in both bars simultaneously, and the lateral dynamic stress follows the same two-bar averaging principle as the X-axis three-wave method [6, 7]:

$$\sigma_y^{\text{dyn}}(t) = \frac{E_b A_b}{2A_s} [\varepsilon_{y1}(t) + \varepsilon_{y2}(t)], \quad (2.26)$$

$$\sigma_z^{\text{dyn}}(t) = \frac{E_b A_b}{2A_s} [\varepsilon_{z1}(t) + \varepsilon_{z2}(t)], \quad (2.27)$$

where $\varepsilon_{y1}, \varepsilon_{y2}$ are the strain-gauge signals on the two Y-axis output bars (faces +Y and −Y), and $\varepsilon_{z1}, \varepsilon_{z2}$ likewise on the two Z-axis bars. The factor of 1/2 in the prefactor is identical to that of the three-wave method (2.15): it is the ratio of *average* face force to specimen cross-sectional area, with both opposing faces contributing equally.

The associated lateral strain histories are obtained by integrating the sum of the two opposing-face velocities, $v_{yi} = C_0 \varepsilon_{yi}$, and dividing by the lateral specimen dimension:

$$\varepsilon_y(t) = \frac{C_0}{L_s} \int_0^t [\varepsilon_{y1}(\tau) + \varepsilon_{y2}(\tau)] d\tau, \quad \varepsilon_z(t) = \frac{C_0}{L_s} \int_0^t [\varepsilon_{z1}(\tau) + \varepsilon_{z2}(\tau)] d\tau. \quad (2.28)$$

For Mode 1 (gas-gun unconfined) the lateral bars are not in contact, and $\sigma_y^{\text{dyn}} = \sigma_z^{\text{dyn}} = 0$ throughout. For Mode 2 (confinement chamber, Chapter 4) the lateral bars are replaced by a pressurised passive chamber, and the lateral stress is held at the constant value $\sigma_y = \sigma_z = p_c$ for the entire test; no lateral wave measurement is available.

To leading order, before damage initiates, the lateral dynamic strain in the active-bar modes is governed by Poisson's ratio:

$$\varepsilon_y^{\text{dyn}}(t) \approx -\nu \varepsilon_x^{\text{dyn}}(t), \quad (2.29)$$

augmented by damage-induced dilation once microcracking initiates (§12.3).

Remark 2.1. Equations (2.26)–(2.28) restore the two-bar averaging factor that was inadvertently dropped from an earlier draft of this handbook. The corrected form is consistent with eqs. (8)–(11) of [6] for the Monash Tri-HB system, and with the original two-output-bar implementation of Cadoni and Albertini (2011).

2.8 Elastic moduli used in the analysis

For completeness the elastic constants of the specimen are summarised. With E_s the Young's modulus and ν the Poisson's ratio,

$$G = \frac{E_s}{2(1+\nu)}, \quad K = \frac{E_s}{3(1-2\nu)}, \quad (2.30)$$

$$M = \frac{E_s(1-\nu)}{(1+\nu)(1-2\nu)}, \quad \lambda = \frac{E_s\nu}{(1+\nu)(1-2\nu)}, \quad (2.31)$$

where G is the shear modulus, K the bulk modulus, M the constrained (P-wave) modulus, and λ the Lamé first parameter. The corresponding specimen wave speeds are

$$c_p = \sqrt{\frac{M}{\rho_s}}, \quad c_s = \sqrt{\frac{G}{\rho_s}}. \quad (2.32)$$

For granite ($E_s = 50$ GPa, $\nu = 0.25$, $\rho_s = 2700$ kg/m³) these give $M = 60$ GPa, $c_p \approx 4714$ m/s, $c_s \approx 2722$ m/s.

2.9 Bar-wave energy partitioning

The energy carried by an elastic wave of strain ε travelling in a bar of cross-section A_b , modulus E_b , and wave speed C_0 is

$$W(t) = E_b A_b C_0 \int_0^t \varepsilon^2(\tau) d\tau, \quad (2.33)$$

which follows from integrating the product of axial force and particle velocity, $F v = E_b A_b \varepsilon \cdot C_0 \varepsilon$, over time. The energy balance at the specimen is

$$W_I = W_R + W_T + W_{\text{abs}}, \quad (2.34)$$

where W_{abs} is the energy absorbed by the specimen (elastic storage plus damage dissipation plus heat).

In the symmetric mode (Mode 6), opposing bar pairs contribute additively:

$$W_I = E_b A_b C_0 \int [(\varepsilon_I^+)^2 + (\varepsilon_I^-)^2] dt, \quad W_R = E_b A_b C_0 \int [(\varepsilon_R^+)^2 + (\varepsilon_R^-)^2] dt. \quad (2.35)$$

The unit check is $A_b [\text{m}^2] \cdot E_b [\text{Pa}] \cdot C_0 [\text{m/s}] \cdot \int \varepsilon^2 dt [\text{s}] = \text{J}$.

The next six chapters apply these foundations to the six loading modes in turn.

Mode 1: Gas-gun uniaxial impact (classical SHPB)

The gas-gun mode is the historical reference configuration of split-Hopkinson testing. A projectile (the striker bar) is launched by compressed gas, impacts the end of the incident bar, and generates a compressive pulse that propagates into the specimen. The mode is treated first because its analysis is the simplest, and because the remaining modes can be understood as generalisations of it.

Mode 1: Gas-gun uniaxial SHPB

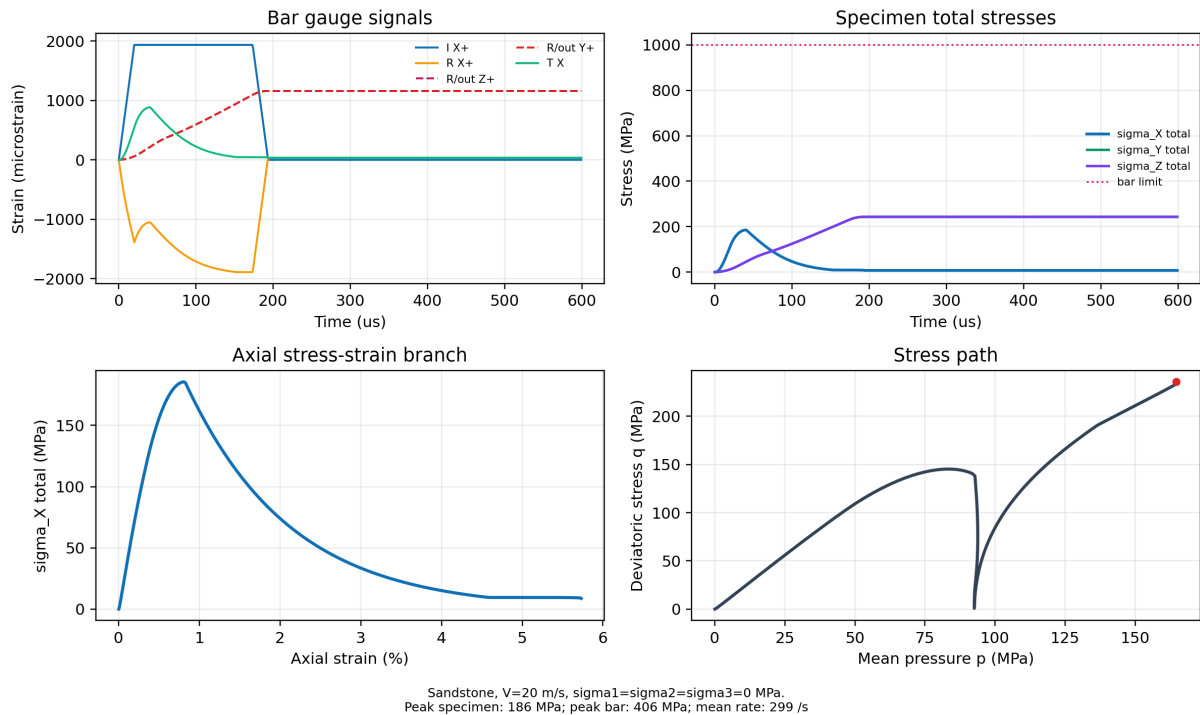


Figure 3.1: Mode 1 sample run: sandstone, gas-gun uniaxial SHPB, $V = 20$ m/s, no static confinement.

3.1 Hardware configuration

- **Striker bar.** 42CrMo steel, length $L_{\text{striker}} = 0.5$ m, impact velocity up to $V = 50$ m/s.
- **Incident and transmission bars.** The Step 1 sidebar of the digital twin offers Square 50×50 mm ($A_b = 2500$ mm²) or Round with an editable diameter, $A_b = \pi(d/2)^2$. The current digital twin uses a default round diameter of $\Phi 50$ ($A_b \approx 1963$ mm²) and defaults to the square cross-section so that the same bars are shared with the electromagnetic bench.

- **Absorption bar and moment-trap.** Prevent end-reflections and tensile contamination of the specimen response.
- **Static pre-stress.** Hydraulic cylinders independently apply σ_1 on X, σ_2 on Y, and σ_3 on Z, slider range 0 MPa to 50 MPa per axis. See §2.5.

3.2 Generation of the incident pulse

When a striker of velocity V impacts a stationary incident bar, both bar ends instantaneously acquire a common particle velocity $V/2$ by momentum and velocity continuity. The compressive stress generated in the incident bar follows from the Hopkinson identity (2.5):

$$\sigma_{\text{peak}} = \frac{1}{2} \rho_b C_0 V = \frac{1}{2} \frac{E_b}{C_0} V. \quad (3.1)$$

The two forms are identical because $E_b = \rho_b C_0^2$; the digital twin internally uses $\frac{1}{2} E_b V / C_0$ so that the editable bar parameters propagate directly to the peak stress.

The pulse duration is set by the round-trip time within the striker bar:

$$\tau = \frac{2L_{\text{striker}}}{C_0}. \quad (3.2)$$

Worked example: striker velocity $V = 20$ m/s

$$\begin{aligned} \sigma_{\text{peak}} &= \frac{1}{2} \left(\frac{210 \text{ GPa}}{5172 \text{ m/s}} \right) (20 \text{ m/s}) \approx 406 \text{ MPa}, \\ \tau &= \frac{2 \times 0.5 \text{ m}}{5172 \text{ m/s}} \approx 193 \mu\text{s}. \end{aligned}$$

A 20 m/s impact thus produces an approximately rectangular pulse of 406 MPa amplitude lasting 193 μs .

3.2.1 The pulse shaper

A thin disc of soft metal (the pulse shaper, typically annealed copper or rubber, 0.5 mm to 2 mm thick) is placed at the impact face. Its plastic deformation smooths the leading edge of the pulse from a near-instantaneous step to a finite rise of 20 μs to 40 μs , which (i) reduces the high-frequency content responsible for Pochhammer–Chree oscillations (§2.1.3), and (ii) gives the specimen time to reach dynamic stress equilibrium before failure. Design rules for pulse shapers are discussed by Frew & Forrestal [11]. The digital twin approximates the shaped pulse with a trapezoidal envelope having 20 μs rise/fall and a flat plateau between.

3.3 Pulse propagation and specimen response

The pulse travels along the incident bar at C_0 . When it reaches the specimen face, three coupled phenomena occur simultaneously:

1. part of the pulse is reflected back into the incident bar as $\varepsilon_R(t)$, normally tensile for brittle specimens;

2. part is transmitted through the specimen into the transmission bar as $\varepsilon_T(t)$, compressive;
3. the specimen deforms at strain rate $\dot{\varepsilon}_s = (C_0/L_s)(\varepsilon_I - \varepsilon_R - \varepsilon_T)$.

The lateral output bars on Y and Z carry only the specimen's lateral expansion (Poisson plus damage dilation), since no active dynamic pulse is applied to those axes in gas-gun mode.

3.4 Reading the strain gauge records

Strain gauges on the incident bar record both ε_I (passing on the way in) and ε_R (passing on the way back). A second pair of gauges on the transmission bar records ε_T . To separate the incident from the reflected signal in time, the gauge is placed at a distance L_g from the specimen such that

$$\frac{L_g}{C_0} > \frac{\tau}{2}, \quad (3.3)$$

guaranteeing that no overlap occurs. For $\tau = 193 \mu\text{s}$ and a typical $L_g = 1.0 \text{ m}$, the round-trip time is $387 \mu\text{s}$, comfortably above the threshold.

3.5 Wave analysis: the three-wave method applied

Equations (2.15)–(2.17) apply directly:

$$\begin{aligned} \sigma_x(t) &= \frac{E_b A_b}{2A_s} [\varepsilon_I(t) + \varepsilon_R(t) + \varepsilon_T(t)], \\ \dot{\varepsilon}_x(t) &= \frac{C_0}{L_s} [\varepsilon_I(t) - \varepsilon_R(t) - \varepsilon_T(t)], \\ \varepsilon_x(t) &= \int_0^t \dot{\varepsilon}_x(\tau) d\tau. \end{aligned}$$

The output is the dynamic stress–strain curve $\sigma_x(\varepsilon_x)$, which is the primary deliverable of the test. The mean strain rate $\bar{\dot{\varepsilon}}$ is taken over the equilibrated portion of the loading branch, typically from $\varepsilon = 0.2\%$ to $\varepsilon = 0.8\%$ for a brittle rock.

3.6 Lateral response on Y and Z

Because no active pulse is applied to the lateral axes, the lateral bars record only the specimen's outward push. Each lateral bar carries a single outgoing wave $\varepsilon_Y(t)$ or $\varepsilon_Z(t)$, and the lateral dynamic stress is given by (2.26)–(2.27). The total lateral stress on each axis is the sum of the static confining pre-stress and the dynamic contribution.

3.7 Validity envelope

- **Striker velocity:** $V \leq 50 \text{ m/s}$ (slider limit). At $V \approx 50 \text{ m/s}$, $\sigma_{\text{peak}} \approx 1020 \text{ MPa}$ slightly exceeds the current $\sigma_{\text{prop}} = 1000 \text{ MPa}$ warning threshold. Tests intended to stay fully elastic should use $V \lesssim 49 \text{ m/s}$, or about 40 m/s for conservative margin.
- **Pre-stress range:** $0 \leq \sigma_1, \sigma_2, \sigma_3 \leq 50 \text{ MPa}$ per axis in the current digital-twin calibration.
- **Strain rate range:** $\dot{\varepsilon} \sim 50 \text{ s}^{-1}$ to 500 s^{-1} .

Mode 2: Confinement-chamber SHPB

The confinement-chamber SHPB is the most widely deployed triaxial-SHPB configuration in the dynamic rock and concrete literature [13, 17, 4, 21, 5]. A cylindrical specimen is held inside a stiff confining chamber — typically an oil-filled pressure vessel sealed by an elastomeric jacket, or a confining sleeve of metal, PU, or epoxy — that delivers an axisymmetric lateral pressure $\sigma_2 = \sigma_3 = p_c$. The axial baseline σ_1 is applied through a hydraulic cylinder acting along the X (loading) line, as in the Monash Tri-HB (Chapter 5), while the dynamic pulse is launched by the same gas-gun bench as in Modes 1 and 3.

The defining contrast with Mode 3 is the *passive* character of the lateral boundary: the confining medium holds p_c constant throughout the dynamic event, and there are no lateral bars from which lateral wave data can be inferred. The arrangement is mechanically simpler than the six-bar Monash architecture, easier to seal against high p_c , and well suited to applications where the axisymmetric stress state is adequate. It cannot, however, deliver independent σ_2 and σ_3 or probe non-axisymmetric stress states: the Lode angle is fixed at $\theta = 0$ (triaxial compression) for the entire test.

Mode 2: Confinement-chamber triaxial SHPB

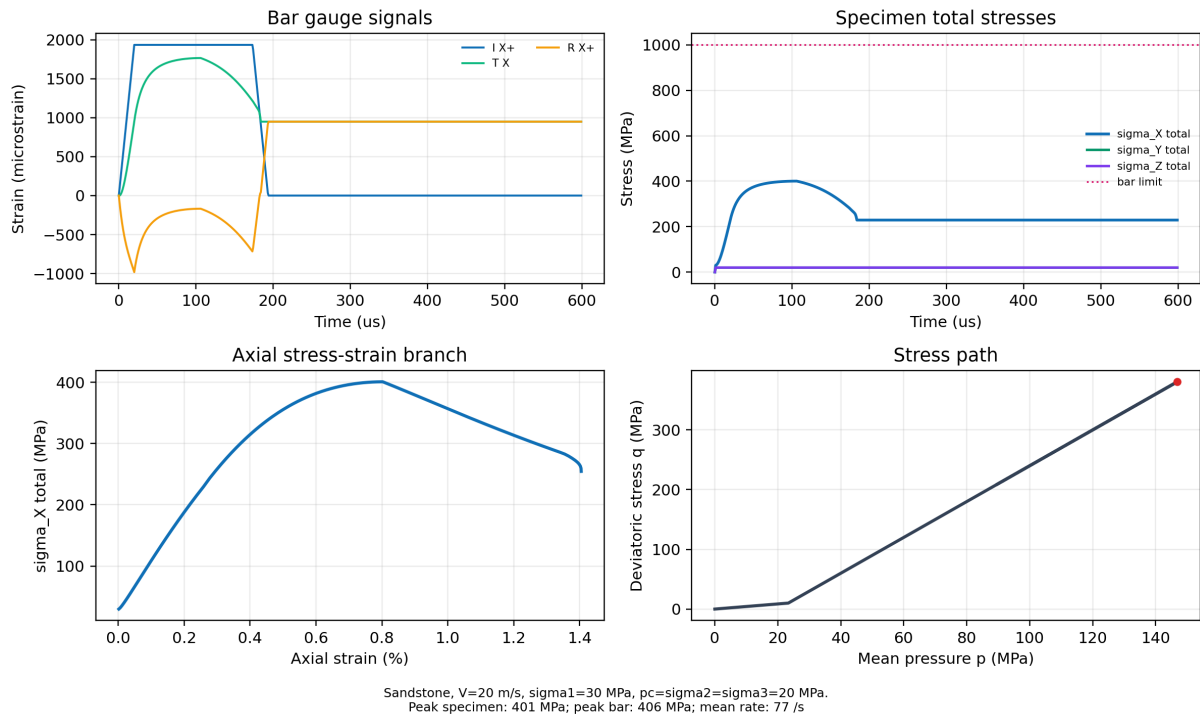


Figure 4.1: Mode 2 sample run: sandstone in a confinement chamber, $V = 20$ m/s, $\sigma_1 = 30$ MPa and $p_c = 20$ MPa.

4.1 Hardware configuration

- **Specimen geometry.** Cylindrical, with the standard aspect ratio $L_s/d_s \in [0.5, 1.0]$ (§2.3.3). Specimen diameter is matched to the bar diameter for the round-bar configuration, or chosen with a small clearance for the confining jacket.
- **X-axis gas-gun train.** Striker bar (0.5 m standard or 1.5 m extended), incident bar, transmission bar, and absorption bar — geometrically identical to the X axis of Mode 1 and Mode 3.
- **X-axis hydraulic cylinder.** Applies the static axial baseline σ_1 , in the range 0 MPa to 50 MPa (matched to the Monash bench rating). The cylinder reaches equilibrium before the gas gun fires; this is the *same* axial loading arrangement as in Mode 3.
- **Confining chamber.** A pressure vessel surrounding the specimen, sealed by an elastomeric jacket and pressurised with hydraulic fluid to p_c , in the range 0 MPa to 50 MPa (slider cap matched to other modes). The chamber wall is stiff steel, much stiffer than the specimen so that radial expansion of the specimen produces negligible drop in p_c during the dynamic event.
- **Lateral bars: absent.** The chamber wall replaces the bar pairs on the Y and Z axes. No lateral wave measurement is available; the lateral stress is held at p_c throughout the event by the chamber.

4.2 Loading sequence

A Mode 2 test proceeds in three sequential phases:

1. **Chamber pressurisation.** The confining fluid is pumped to the target p_c at a rate slow enough that no significant wave is launched in the specimen (30 s to 60 s). The specimen experiences a hydrostatic compression of magnitude p_c .
2. **Axial pre-loading.** The X-axis hydraulic cylinder ramps the axial baseline σ_1 to its target value, again slowly, producing an additional axial compression of magnitude $\sigma_1 - p_c$ relative to the hydrostatic baseline (so that the total axial stress is σ_1 and the lateral is p_c).
3. **Dynamic pulse.** The gas gun fires the striker bar at velocity V , producing the standard Hopkinson incident pulse on the incident bar. The dynamic increment σ_X^{dyn} adds to the axial baseline; the lateral stress remains at p_c throughout (subject to small chamber-compliance corrections, §4.6).

The total stress history during the dynamic event is therefore

$$\begin{cases} \sigma_X^{\text{total}}(t) = \sigma_1 + \sigma_X^{\text{dyn}}(t), \\ \sigma_Y^{\text{total}}(t) = p_c, \\ \sigma_Z^{\text{total}}(t) = p_c. \end{cases} \quad (4.1)$$

The lateral stress is constant in time (to leading order); only the X axis carries dynamic information.

4.3 Wave analysis on the active (X) axis

The X-axis bars carry incident, reflected, and transmitted waves identically to Modes 1 and 3. The three-wave equations of §2.3 apply unchanged:

$$\begin{aligned}\sigma_X^{\text{dyn}}(t) &= \frac{E_b A_b}{2A_s} [\varepsilon_I(t) + \varepsilon_R(t) + \varepsilon_T(t)], \\ \dot{\varepsilon}_X(t) &= \frac{C_0}{L_s} [\varepsilon_I(t) - \varepsilon_R(t) - \varepsilon_T(t)], \\ \varepsilon_X(t) &= \int_0^t \dot{\varepsilon}_X(\tau) d\tau.\end{aligned}$$

With a cylindrical specimen and round bars, $A_s = \pi d_s^2/4$ and $A_b = \pi d_b^2/4$; for matched diameters, $A_s = A_b$.

4.4 Stress invariants and Lode angle

Because $\sigma_2 = \sigma_3 = p_c$ throughout the test (axisymmetric), the stress invariants collapse to one-parameter families of σ_X :

$$p(t) = \frac{1}{3} [\sigma_X^{\text{total}}(t) + 2p_c], \quad (4.2)$$

$$q(t) = |\sigma_X^{\text{total}}(t) - p_c|, \quad (4.3)$$

$$\theta = 0 \quad (\text{triaxial compression, fixed}). \quad (4.4)$$

A Mode 2 test therefore traces a single radial line in p - q space, parameterised by p_c . To explore a different Lode angle requires a different test (true triaxial, e.g. Mode 3 or Mode 5), since the confinement chamber cannot break the $\sigma_2 = \sigma_3$ symmetry.

4.5 Confinement strengthening

The Mohr–Coulomb-type strengthening relation (2.24) applies with $\max(\sigma_2, \sigma_3) = p_c$:

$$\sigma_{\text{peak}}^{\text{rock,dyn}} = (\sigma_{c0} + k_{\text{conf}} p_c) \cdot \text{DIF}, \quad (4.5)$$

the same planning relation that applies to Mode 3 with $\sigma_2 = \sigma_3$ specialised to the hydrostatic case. The axial baseline σ_1 raises the total axial stress at peak by a constant offset but does not affect the dynamic peak strength of the rock itself.

4.6 Two parasitic corrections

The simplicity of the chamber configuration comes at the price of two small-but-real corrections that are absent from the active-bar modes.

4.6.1 Chamber compliance

A finite-stiffness chamber expands slightly when the specimen pushes outward during the dynamic event, so the effective lateral pressure deviates from the nominal p_c . To leading order,

$$p_c^{\text{eff}}(t) \approx p_c + K_c \varepsilon_{r,s}(t), \quad (4.6)$$

where $\varepsilon_{r,s}(t)$ is the radial (lateral) strain of the specimen and K_c is the effective radial stiffness of the chamber wall per unit specimen lateral strain. For thick steel chambers K_c is high enough that this correction is below 1 % and is normally neglected; for thin sleeves or fluid-only confinement, it can reach several percent and should be calibrated against quasi-static expansion tests of the empty chamber.

4.6.2 Jacket parasitic load

If the elastomeric jacket or metal sleeve carries a portion of the axial load in parallel with the specimen, the transmission-bar reading overestimates the specimen force. The correction is

$$F_s(t) = F_{\text{bar}}(t) - A_j E_j \varepsilon_X(t), \quad (4.7)$$

where A_j and E_j are the jacket cross-section and modulus. Soft elastomer jackets ($E_j \sim 10$ MPa) contribute a negligible correction; metal jackets ($E_j \sim 200$ GPa) require explicit subtraction.

When the corrections matter

Steel chamber + elastomer jacket (the most common arrangement): both corrections are negligible; the bare three-wave equations are accurate.

Thin-walled sleeve confinement (low-cost alternative): chamber compliance can be 5 % to 15 % and must be calibrated.

Metal-jacketed specimen (used for very weak materials, e.g. soils): jacket parasitic load must be subtracted; otherwise the apparent dynamic strength is overestimated.

4.7 Procedure

Mode 2 procedure

1. Mount the cylindrical specimen with its jacket inside the confining chamber. Verify alignment of the chamber with the incident and transmission bar ends.
2. Pressurise the confining chamber slowly to the target p_c (default $p_c = 20$ MPa). Allow the system to settle (30 s to 60 s).
3. Ramp the X-axis hydraulic cylinder to the target axial baseline σ_1 (default $\sigma_1 = 30$ MPa). Allow the system to settle.
4. Record the pre-trigger baseline on the incident, reflected, and transmitted bar gauges.
5. Choose the striker velocity V to deliver the desired $\sigma_{\text{peak}} = \frac{1}{2}\rho_b C_0 V$, using (4.5) as the planning relation. Verify $\sigma_{\text{peak}} < \sigma_{\text{prop}} = 1000$ MPa.
6. Trigger the gas gun. Capture all X-axis gauge signals at ≥ 1 MHz. Monitor the chamber pressure transducer through the event.
7. Subtract the pre-trigger baselines.
8. Time-shift each channel to the specimen-face reference.
9. Verify stress equilibrium $\mathcal{R}(t) < 0.05$ ((2.19)).
10. Apply the three-wave equations to obtain $\sigma_X^{\text{dyn}}(t)$, $\dot{\varepsilon}_X(t)$, $\varepsilon_X(t)$.

11. Apply the chamber-compliance correction (4.6) if the chamber stiffness warrants it; apply the jacket-load correction (4.7) if a metal jacket is used.
12. Reconstruct the total stresses per (4.1).
13. Plot σ_X^{total} against ε_X , extract peak strength and mean strain rate. Compare with the planning prediction (4.5).
14. Confirm the maximum bar stress remained below σ_{prop} .

4.8 Worked example: sandstone in a steel chamber

A sandstone specimen ($\sigma_{c0} = 80$ MPa, $k_{\text{conf}} = 4.5$, $b_{\text{rate}} = 0.18$) is to be tested at the digital-twin default state for the Confinement-chamber mode: $\sigma_1 = 30$ MPa axial, $p_c = \sigma_2 = \sigma_3 = 20$ MPa confining, with a striker velocity $V = 20$ m/s.

Step 1. Incident pulse amplitude:

$$\sigma_{\text{peak}} = \frac{1}{2}\rho_b C_0 V \approx 406 \text{ MPa}.$$

Step 2. Dynamic peak rock strength from (4.5), assuming $\bar{\varepsilon} \approx 150 \text{ s}^{-1}$:

$$\begin{aligned} \text{DIF} &= 1 + 0.18 \log_{10}\left(\frac{150}{10^{-5}}\right) \approx 2.29, \\ \sigma_{\text{peak}}^{\text{rock,dyn}} &= (80 + 4.5 \times 20) \cdot 2.29 \approx 390 \text{ MPa}. \end{aligned}$$

Step 3. Total axial stress at peak:

$$\sigma_{\text{peak}}^{\text{total}} = \sigma_1 + \sigma_{\text{peak}}^{\text{rock,dyn}} = 30 + 390 \approx 420 \text{ MPa}.$$

Step 4. Stress invariants at peak (axisymmetric, $\theta = 0$):

$$p = \frac{1}{3}(420 + 2 \times 20) \approx 153 \text{ MPa}, \quad q = |420 - 20| \approx 400 \text{ MPa}.$$

Step 5. Bar-stress check:

$$\sigma_{\text{bar}} \approx 390 \text{ MPa} < \sigma_{\text{prop}} = 1000 \text{ MPa} \checkmark.$$

Comment. The dynamic peak rock strength at $p_c = 20$ MPa is identical to that computed for Mode 3 in the worked example below (Chapter 5) at the same $\max(\sigma_2, \sigma_3) = 20$ MPa. The two modes therefore give equivalent peak-strength predictions when the Monash Tri-HB happens to be operated with $\sigma_2 = \sigma_3$. What Mode 2 *cannot* reproduce is the response under $\sigma_2 \neq \sigma_3$ (e.g., the default Mode 3 state $\sigma_2 = 20$, $\sigma_3 = 15$ with $\theta > 0$): for that, the active-bar architecture of Mode 3 is required.

4.9 Validity envelope

- **Striker velocity:** the Streamlit slider permits $V \leq 50$ m/s; the bar-limit warning should remain clear, so $V \lesssim 49$ m/s is the practical elastic upper bound for the default bar constants.

- **Confining pressure:** $0 \leq p_c \leq 50$ MPa for the digital-twin slider; the physical chamber rating may permit higher values (100 MPa to 300 MPa in published rigs).
- **Axial baseline:** $0 \leq \sigma_1 \leq 50$ MPa.
- **Strain rate range:** $\dot{\epsilon} \sim 50 \text{ s}^{-1}$ to 500 s^{-1} .
- **Lode angle:** fixed at $\theta = 0$. For research questions requiring $\theta > 0$, use Mode 3 or Mode 5.

Mode 3: Monash Tri-HB — gas-gun coupled static–dynamic triaxial loading

This mode is the signature physical capability of the Monash Tri-HB facility (§1.1). The gas-gun hardware of Mode 1 is retained, but active triaxial confining pre-stress is applied on the Y and Z axes through the lateral hydraulic cylinders *before* the striker bar fires. The specimen therefore experiences a quasi-static triaxial stress state plus a uniaxial dynamic increment along X — the loading regime representative of an in-situ rockmass struck by a blast or seismic wave [6, 7].

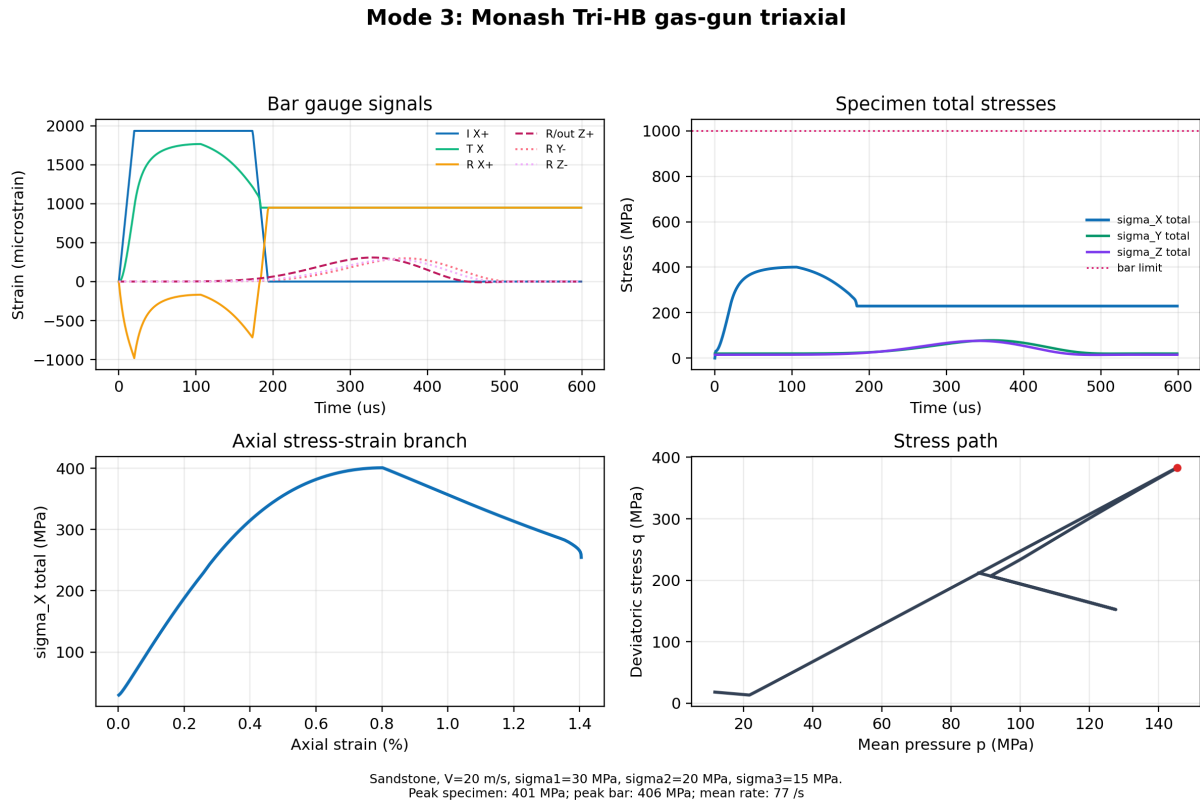


Figure 5.1: Mode 3 sample run: sandstone with independent static triaxial pre-stress and gas-gun dynamic loading, $V = 20$ m/s.

5.1 Hardware configuration

The bench and bar geometry are identical to Mode 1: striker (0.5 m for the standard configuration; 1.5 m for the Monash extended striker), incident bar (2.5 m), transmission bar (2 m), and

absorption bar (0.5 m) along X. What distinguishes Mode 3 from Mode 1 is the *active* role of the lateral train:

- **Y-axis bars** ($\pm Y$, four bars of 2 m) loaded by a horizontal hydraulic cylinder to apply σ_2 on the specimen.
- **Z-axis bars** ($\pm Z$, four bars of 2 m) loaded by a vertical hydraulic cylinder to apply σ_3 on the specimen.
- **X-axis hydraulic cylinder** optionally applies an axial baseline σ_1 .
- **Reaction frames** (six high-strength steel) carry the static pre-load and isolate the gas-gun bench from the cylinder line of action.

The four lateral output bars (one bonded to each of the four lateral faces of the cubic specimen: $\pm Y$ and $\pm Z$) record the specimen’s lateral wave response in real time, allowing the dynamic Y and Z stresses to be inferred from (2.26)–(2.27). The hydraulic cylinders are servo-controlled and reach equilibrium with the specimen before the gas gun is triggered.

5.2 Loading sequence and superposition

A Mode 3 test proceeds in two distinct phases:

1. **Quasi-static pre-loading.** The hydraulic cylinders ramp $\sigma_1, \sigma_2, \sigma_3$ to their target values over 30 s to 60 s, slow enough that the bars deform elastically and no stress wave is generated. The strain gauges record this as a low-frequency baseline.
2. **Dynamic pulse.** The gas gun fires the striker bar at velocity V , producing the standard Hopkinson incident pulse $\sigma_{\text{peak}} = \frac{1}{2}\rho_b C_0 V$ on the incident bar. The dynamic pulse reaches the specimen and is processed by the three-wave method exactly as in Mode 1.

Because the two phases are temporally separated, the specimen-stress response is the linear superposition of the static pre-stress and the dynamic increment, per (2.23):

$$\begin{aligned}\sigma_X^{\text{total}}(t) &= \sigma_1 + \sigma_X^{\text{dyn}}(t), \\ \sigma_Y^{\text{total}}(t) &= \sigma_2 + \sigma_Y^{\text{dyn}}(t), \\ \sigma_Z^{\text{total}}(t) &= \sigma_3 + \sigma_Z^{\text{dyn}}(t).\end{aligned}\tag{5.1}$$

On the X axis σ_X^{dyn} is the active dynamic stress driven by the striker; on Y and Z, σ_Y^{dyn} and σ_Z^{dyn} are the small Poisson-coupled lateral responses $\sigma_{\text{lat}}^{\text{dyn}} \approx -\nu\sigma_X^{\text{dyn}}$ measured on the lateral output bars.

5.2.1 Why the bar gauges do not record the pre-stress

As discussed in §2.5, the static pre-stress is held by the hydraulic cylinders and is subtracted as the pre-event baseline at data acquisition. The wave-analysis equations operate exclusively on the dynamic component; the static pre-stress is reintroduced at the reporting stage through the superposition above. This is the same convention used in Modes 1 and 3–5.

5.3 Wave analysis on the active (X) axis

The three-wave equations of §2.3 apply unchanged:

$$\begin{aligned}\sigma_X^{\text{dyn}}(t) &= \frac{E_b A_b}{2A_s} [\varepsilon_I(t) + \varepsilon_R(t) + \varepsilon_T(t)], \\ \dot{\varepsilon}_X(t) &= \frac{C_0}{L_s} [\varepsilon_I(t) - \varepsilon_R(t) - \varepsilon_T(t)], \\ \varepsilon_X(t) &= \int_0^t \dot{\varepsilon}_X(\tau) d\tau.\end{aligned}$$

The bars remain the diagnostic instruments; the constitutive interpretation of $(\sigma_X^{\text{total}}, \varepsilon_X)$ must now account for the confining pre-stress on Y and Z via a multiaxial failure surface (Chapter 12).

5.4 Confinement strengthening

The principal value of Mode 3 over Mode 1 is the strengthening induced by active confinement. For low-to-moderate confining pressure, the dynamic peak rock strength follows the Mohr–Coulomb-type relation (2.24):

$$\sigma_{\text{peak}}^{\text{rock,dyn}} = [\sigma_{c0} + k_{\text{conf}} \max(\sigma_2, \sigma_3)] \cdot \text{DIF}, \quad (5.2)$$

where k_{conf} is the confinement coefficient and DIF the rate factor of (2.25). This is the equation underpinning the Liu et al. [6] sandstone dataset and forms the planning relation for selecting striker velocity and pre-stress combinations.

Why σ_2 and σ_3 strengthen but σ_1 does not

The axial baseline σ_1 is collinear with the dynamic pulse: it shifts the total axial stress curve upward without altering the deviatoric stress the specimen must resist. The lateral pre-stresses σ_2 and σ_3 , on the other hand, are perpendicular to the loading axis: they reduce the deviatoric stress at any given σ_X and therefore raise the dynamic peak through the Mohr–Coulomb mechanism. Exercise 7 of Appendix B explores this asymmetry quantitatively.

5.5 Procedure

Mode 3 procedure

1. Place the cubic specimen between the two X bars. Bring the four lateral output bars on Y and Z into contact with the specimen faces; verify alignment.
2. Set the target confining pre-stresses σ_2 and σ_3 on the Y and Z hydraulic cylinders. Optionally set an axial baseline σ_1 on the X cylinder. Ramp slowly (30 s to 60 s) and allow the system to settle.
3. Record the pre-trigger baseline on all bar gauges (incident, reflected, transmitted, and the four lateral output gauges Y_1, Y_2, Z_1, Z_2).
4. Choose the striker velocity V to deliver the desired dynamic peak $\sigma_{\text{peak}} = \frac{1}{2}\rho_b C_0 V$, using (5.2) as the planning relation. Verify $\sigma_{\text{peak}} < \sigma_{\text{prop}} = 1000 \text{ MPa}$.

5. Trigger the gas gun. Capture all gauge signals at sampling rates ≥ 1 MHz.
6. Subtract the pre-trigger baseline from each channel (this removes the static pre-stress from the recorded dynamic signals).
7. Time-shift each channel to the specimen-face reference.
8. Verify stress equilibrium $\mathcal{R}(t) < 0.05$ per (2.19).
9. Apply the three-wave equations to obtain $\sigma_X^{\text{dyn}}(t)$, $\dot{\varepsilon}_X(t)$, and $\varepsilon_X(t)$.
10. From the lateral output bars compute the Poisson-coupled responses $\sigma_Y^{\text{dyn}}(t)$ and $\sigma_Z^{\text{dyn}}(t)$ via (2.26)–(2.27).
11. Add the static pre-stresses back per (2.23) to obtain the total stress histories σ_X^{total} , σ_Y^{total} , σ_Z^{total} .
12. Plot σ_X^{total} against ε_X , extract the peak strength and mean strain rate. Compare with the planning prediction (5.2).
13. Confirm the maximum bar stress remained below σ_{prop} .

5.6 Worked example: sandstone under triaxial pre-stress

A sandstone specimen ($\sigma_{c0} = 80$ MPa, $k_{\text{conf}} = 4.5$, $b_{\text{rate}} = 0.18$) is to be tested at the digital-twin default pre-stress state for the Monash Tri-HB mode, $\sigma_1 = 30$ MPa axial, $\sigma_2 = 20$ MPa, $\sigma_3 = 15$ MPa (note that σ_2 and σ_3 may be set independently in this mode), with a striker velocity $V = 20$ m/s.

Step 1. Incident pulse amplitude from the Hopkinson identity:

$$\sigma_{\text{peak}} = \frac{1}{2} \rho_b C_0 V = \frac{1}{2} (7850)(5172)(20) \cdot 10^{-6} \approx 406 \text{ MPa}.$$

Step 2. Estimated dynamic peak rock strength from (5.2), taking $\max(\sigma_2, \sigma_3) = 20$ MPa and assuming $\bar{\varepsilon} \approx 150 \text{ s}^{-1}$:

$$\begin{aligned} \text{DIF} &= 1 + 0.18 \log_{10} \left(\frac{150}{10^{-5}} \right) \approx 2.29, \\ \sigma_{\text{peak}}^{\text{rock,dyn}} &= (80 + 4.5 \times 20) \cdot 2.29 \approx 390 \text{ MPa}. \end{aligned}$$

Step 3. Total axial stress at peak:

$$\sigma_{\text{peak}}^{\text{total}} = \sigma_1 + \sigma_{\text{peak}}^{\text{rock,dyn}} = 30 + 390 \approx 420 \text{ MPa}.$$

Step 4. Bar-stress check. The bar must transmit a stress equal to $\sigma_{\text{peak}}^{\text{rock,dyn}} \cdot A_s/A_b$. With matched 50 mm cubic specimen and square bars, $A_s/A_b = 1$:

$$\sigma_{\text{bar}} \approx 390 \text{ MPa} < \sigma_{\text{prop}} = 1000 \text{ MPa} \checkmark.$$

Step 5. Comparison with Mode 1 (same striker, no confinement). The unconfined dynamic peak would be only

$$\sigma_{\text{peak},0}^{\text{rock,dyn}} = \sigma_{c0} \cdot \text{DIF} \approx 80 \cdot 2.29 \approx 183 \text{ MPa},$$

a factor of ~ 2.1 below the confined value. This is the dynamic analogue of the well-known confinement strengthening of quasi-static triaxial tests; the Monash Tri-HB allows it to be measured at high strain rate [6].

5.7 Validity envelope

- **Striker velocity:** the Streamlit slider permits $V \leq 50$ m/s; keep the bar-limit warning clear ($V \lesssim 49$ m/s for the default bar constants; see §3.7).
- **Confining pre-stress range:** $0 \leq \sigma_2, \sigma_3 \leq 50$ MPa per axis (slider cap of the current digital-twin calibration; matched to the 50 MPa hydraulic-cylinder rating of the Monash bench).
- **Axial baseline:** $0 \leq \sigma_1 \leq 50$ MPa. The total axial stress $\sigma_1 + \sigma_X^{\text{dyn}}$ must remain within the constitutive validity range of the chosen failure model.
- **Strain rate range:** $\dot{\epsilon} \sim 50 \text{ s}^{-1}$ to 500 s^{-1} .
- **Confinement vs strain rate.** Increasing confinement σ_2, σ_3 raises the dynamic peak strength but also reduces the achieved strain rate at a given striker velocity, because the specimen deforms less before failure. The two parameters should be swept independently for a complete constitutive characterisation.

Mode 4: Electromagnetic uniaxial impact

The electromagnetic uniaxial mode replaces the gas gun and striker bar with a capacitor-bank-driven coil that produces a Lorentz force on a flyer plate (or directly on the bar end), launching a stress pulse into a single incident bar. The data-reduction equations are identical to Mode 1; the distinguishing feature is the pulse shape and the independent control of amplitude and duration.

Mode 4: EM uniaxial half-sine

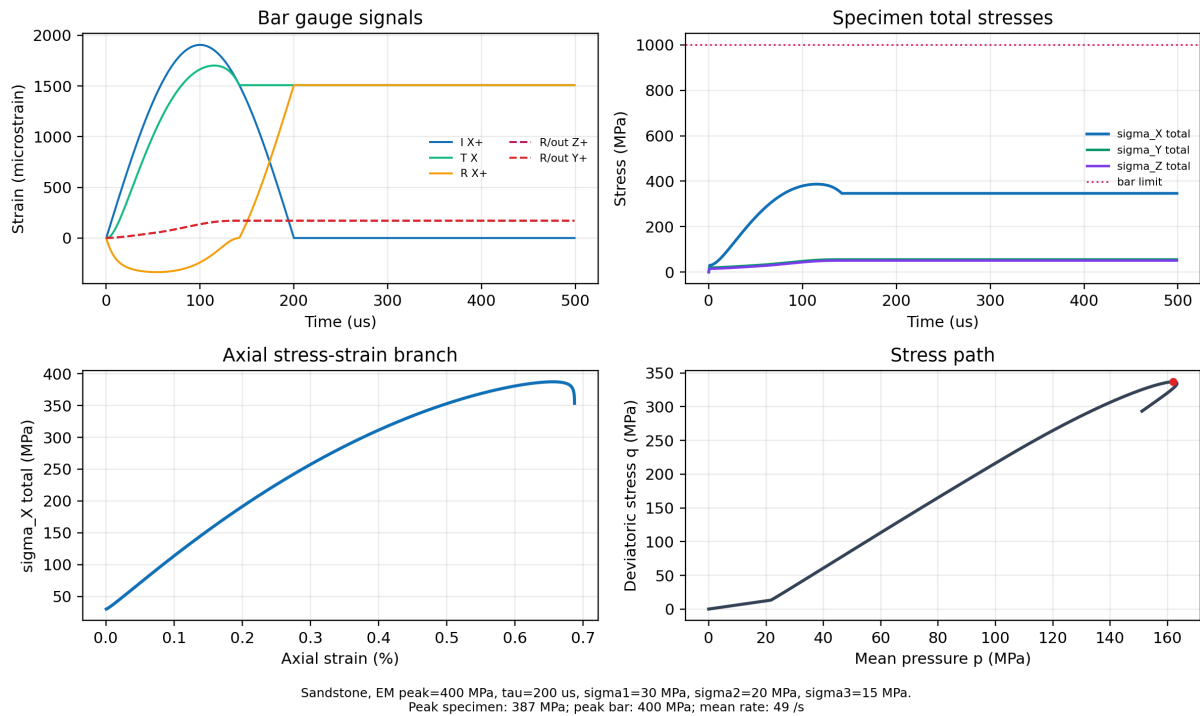


Figure 6.1: Mode 4 sample run: sandstone, single electromagnetic half-sine pulse on +X, $\sigma_{\text{peak}} = 400$ MPa and $\tau = 200$ μs .

6.1 Hardware differences from Mode 1

- **Pulse generator:** capacitor bank discharged through a primary coil produces a Lorentz force on the flyer/bar end.
- **Bars:** square cross-section 50 mm $\times$ 50 mm ($A_b = 2500$ mm 2) by default, for flat contact with cubic specimens. Editable in the Step 1 sidebar.
- **Pre-stress:** same as Mode 1.

6.2 The half-sine pulse

Where the gas-gun pulse is trapezoidal, the electromagnetic generator produces a clean half-sine pulse:

$$\sigma_I(t) = \sigma_{\text{peak}} \sin\left(\frac{\pi t}{\tau}\right), \quad 0 \leq t \leq \tau. \quad (6.1)$$

The peak amplitude σ_{peak} and duration τ are *independently* tuneable through the capacitor charging voltage and the circuit inductance, respectively. This decoupling is the chief advantage of the electromagnetic system: students can sweep the strain-rate axis by varying τ alone, holding σ_{peak} fixed, a controlled experiment that the gas gun cannot provide because there σ_{peak} and τ are linked through the striker geometry and velocity.

6.2.1 Frequency content of the half-sine

A perfectly rectangular pulse contains arbitrarily high-frequency Fourier components, which travel at slightly different speeds in the bar and develop into spurious oscillations as the pulse propagates (§2.1.3). The Fourier transform of the half-sine (6.1) is

$$|\hat{\sigma}_I(\omega)| = \frac{\sigma_{\text{peak}} \pi}{|\omega^2 \tau^2 - \pi^2|} |1 + \cos(\omega \tau)|^{1/2}, \quad (6.2)$$

whose energy is concentrated near the fundamental frequency $f_0 = 1/(2\tau)$ with rapidly decaying higher harmonics. Less high-frequency content means less dispersion and cleaner signals. A half-sine pulse also reaches dynamic stress equilibrium faster than a rectangular pulse of the same amplitude, because its smooth rise gives the specimen time to reverberate internally before peak stress is reached — a property critical for brittle materials.

6.3 Step-by-step procedure

Mode 4 procedure

1. Apply static pre-stress σ_1 (axial), σ_2 , σ_3 (confining) via the three independent servo-cylinders. Record baseline strain on all gauges and let the system settle (30 s to 60 s).
2. Charge the capacitor bank to the voltage required for the desired σ_{peak} , and set the circuit inductance for the desired τ .
3. Trigger the pulse generator. Capture all bar gauge signals at sampling rates ≥ 1 MHz.
4. Subtract the pre-trigger baseline from each channel (this is what removes the static pre-stress from the recorded signal: §2.5).
5. Time-shift each channel to the specimen-face reference: incident gauge forward by L_g/C_0 , reflected backward by L_g/C_0 , etc.
6. Verify stress equilibrium: $\mathcal{R}(t) < 0.05$ per (2.19).
7. Apply the three-wave equations to obtain $\sigma_x^{\text{dyn}}(t)$, $\dot{\epsilon}_x(t)$, $\epsilon_x(t)$.
8. Add the static σ_1 baseline to obtain the total axial stress per (2.23).
9. Plot σ_x^{total} against ϵ_x , extract the peak strength and mean strain rate.
10. Confirm that the maximum bar stress remained below $\sigma_{\text{prop}} = 1000$ MPa.

6.4 Worked example: sandstone with triaxial pre-stress

Consider a sandstone specimen with quasi-static UCS $\sigma_{c0} \approx 80$ MPa, Mohr–Coulomb slope $k_{\text{conf}} \approx 4.5$, and rate-hardening coefficient $b_{\text{rate}} = 0.18$, to be tested at the digital-twin default pre-stress state for the electromagnetic modes, $\sigma_1 = 30$ MPa axial baseline, $\sigma_2 = 20$ MPa, $\sigma_3 = 15$ MPa, at a target strain rate $\dot{\epsilon} \approx 200 \text{ s}^{-1}$.

Step 1. Predicted dynamic peak strength of the specimen, using (2.24) and (2.25), with $\max(\sigma_2, \sigma_3) = 20$ MPa:

$$\begin{aligned} \text{DIF} &= 1 + 0.18 \log_{10} \left(\frac{200}{10^{-5}} \right) \approx 2.31, \\ \sigma_{\text{peak}}^{\text{rock,dyn}} &= (\sigma_{c0} + k_{\text{conf}} \max(\sigma_2, \sigma_3)) \cdot \text{DIF} \\ &= (80 + 4.5 \times 20) \text{ MPa} \cdot 2.31 \approx 393 \text{ MPa}. \end{aligned}$$

Step 2. Total axial stress at peak:

$$\sigma_{\text{peak}}^{\text{total}} = \sigma_1 + \sigma_{\text{peak}}^{\text{rock,dyn}} = 30 + 393 \approx 423 \text{ MPa}.$$

Step 3. Required bar stress. The bar carries only the dynamic component (the static σ_1 being held by the cylinder); with $A_s = A_b$ for a 50 mm cubic specimen between matched square bars,

$$\sigma_{\text{bar}} = \sigma_{\text{peak}}^{\text{rock,dyn}} \cdot \frac{A_s}{A_b} \approx 393 \text{ MPa}.$$

Conclusion. 393 MPa is well below the 1000 MPa bar proportional limit ($\sim 39\%$ of σ_{prop}): safe to proceed. Charge the capacitor for $\sigma_{\text{peak}} \approx 500$ MPa (allowing a margin above 393 MPa to drive the specimen through peak), choose $\tau \approx 200 \mu\text{s}$ to achieve the target strain rate. The digital twin reports a total axial stress reaching ~ 423 MPa on the σ_X trace, comprising the static $\sigma_1 = 30$ MPa plus the dynamic peak.

Mode 5: Electromagnetic asynchronous triaxial impact

This mode uses three of the six electromagnetic generators to apply pulses along the $+X$, $+Y$, and $+Z$ axes, each with an independent time delay. It enables the study of how the *order* of multi-axis loading affects material failure — a question that classical uniaxial Hopkinson testing cannot address.

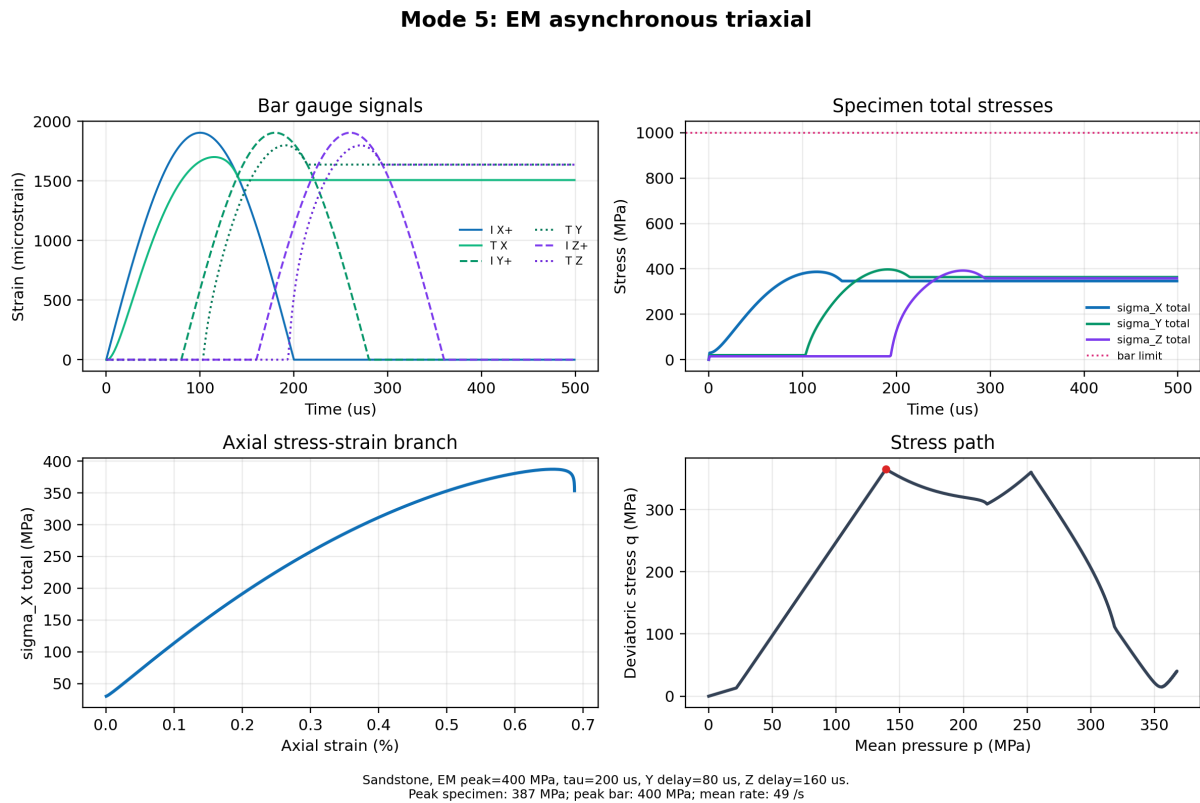


Figure 7.1: Mode 5 sample run: sandstone, electromagnetic pulses on $+X$, $+Y$, and $+Z$ with 80 μ s and 160 μ s inter-axis delays.

7.1 Hardware configuration

Three pulse generators fire on three orthogonal incident bars. The remaining three bars (at the $-X$, $-Y$, $-Z$ ends) act as transmission/output bars and remain passive. In this mode, each of the three positive-axis directions hosts its own three-wave problem; the negative-axis directions show transmitted signals only.

7.2 The time-delay concept

Let pulse $i \in \{x, y, z\}$ be

$$\sigma_{I,i}(t) = \sigma_{\text{peak}} \sin\left(\frac{\pi(t - \Delta t_i)}{\tau}\right), \quad \Delta t_i \leq t \leq \Delta t_i + \tau. \quad (7.1)$$

Setting $\Delta t_x = 0$, $\Delta t_y = 50 \mu\text{s}$, $\Delta t_z = 100 \mu\text{s}$ produces sequential triaxial loading: the specimen first experiences uniaxial dynamic compression along X, then biaxial after 50 μs , then triaxial after 100 μs .

Why ordering matters

Brittle failure in rock and concrete is path-dependent [22]. Cracks initiated in one loading direction can be either suppressed (if confinement arrives quickly) or allowed to propagate (if confinement arrives late). Asynchronous loading isolates this effect from the synchronous case, producing data on the fundamental question of whether dynamic constitutive models require an explicit stress-path term.

Steps 2–3 of the integrated workspace (Chapter 12) turn this informal question into a quantitative one through the normalised-delay parameter $\Delta t^* = \Delta t/t_{\text{travel}}$ and the regime-classifier map of §12.1.

7.3 Three-wave analysis on each axis

Because each axis carries its own incident wave, each axis is analysed with the full three-wave method independently:

$$\sigma_i(t) = \frac{E_b A_b}{2A_s} [\varepsilon_{I,i}(t) + \varepsilon_{R,i}(t) + \varepsilon_{T,i}(t)], \quad i \in \{x, y, z\}, \quad (7.2)$$

$$\dot{\varepsilon}_i(t) = \frac{C_0}{L_s} [\varepsilon_{I,i}(t) - \varepsilon_{R,i}(t) - \varepsilon_{T,i}(t)], \quad i \in \{x, y, z\}. \quad (7.3)$$

The full dynamic stress state of the specimen is the diagonal tensor

$$\boldsymbol{\sigma}(t) = \begin{bmatrix} \sigma_1^0 + \sigma_x^{\text{dyn}}(t) & 0 & 0 \\ 0 & \sigma_2^0 + \sigma_y^{\text{dyn}}(t) & 0 \\ 0 & 0 & \sigma_3^0 + \sigma_z^{\text{dyn}}(t) \end{bmatrix}, \quad (7.4)$$

where the superscript 0 denotes the static pre-stress applied before the dynamic event.

7.4 Procedure

Mode 5 procedure

1. Apply static pre-stress $\sigma_1^0, \sigma_2^0, \sigma_3^0$. Record baselines.
2. Choose pulse delays $\Delta t_y, \Delta t_z$ relative to $\Delta t_x = 0$. Typical values: $\Delta t = 0$ (synchronous, ground truth), $\Delta t = \tau/4$ (mild asynchrony), $\Delta t = \tau/2$ (strong asynchrony).
3. Charge all three capacitor banks to identical voltages for equal σ_{peak} on each axis.

4. Trigger generators with the programmed delays. Synchronisation jitter must be much less than τ ; aim for $< 1 \mu\text{s}$ on a $200 \mu\text{s}$ pulse.
5. Capture each active axis as its own one-sided three-wave record: incident, reflected, and transmitted/output histories on X, Y, and Z. In the current Streamlit implementation the $-Y$ and $-Z$ bars are the transmitted/output bars for their axes; there are no additional passive lateral-output gauges beyond those per-axis three-wave records.
6. Subtract baselines, time-align each channel to its own specimen face.
7. Apply the three-wave method on each axis independently.
8. Plot the principal-stress trajectory $(\sigma_1(t), \sigma_2(t), \sigma_3(t))$ in principal-stress space (or use the p - q projection of Chapter 12).
9. Compare against a matched synchronous test ($\Delta t = 0$) to extract the path-dependence effect.

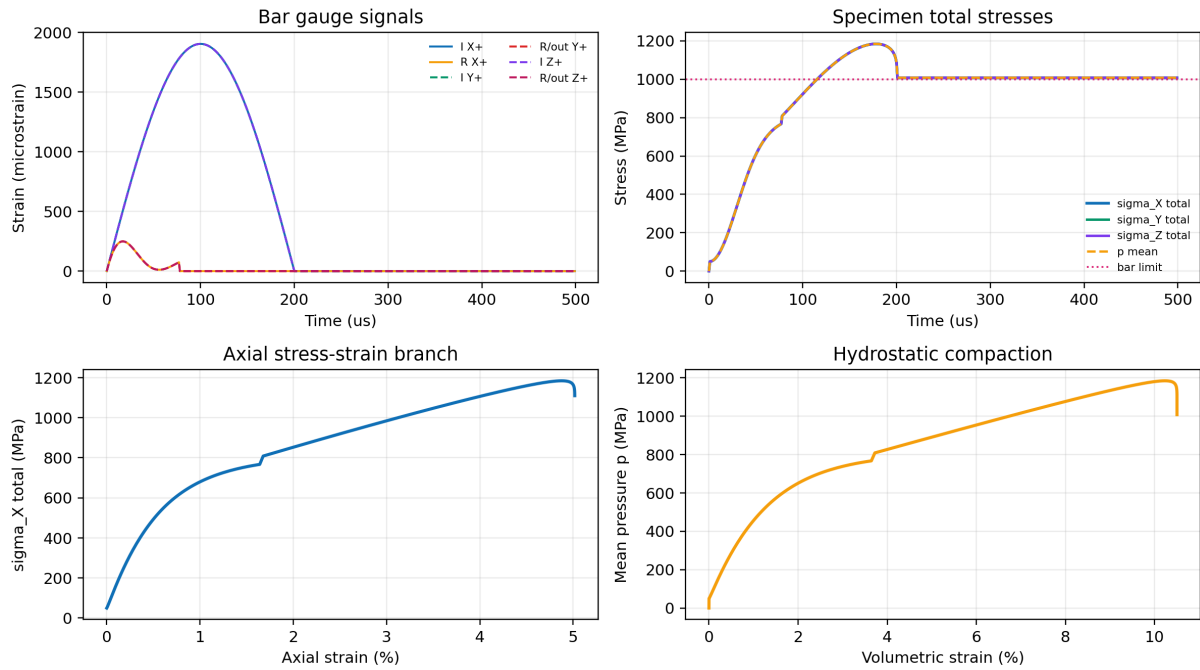
7.5 Validity notes

- All three axes are subject to the same bar plasticity limit. For $\sigma_{\text{peak}} = 500 \text{ MPa}$ pulses on all three axes, no individual bar exceeds 500 MPa.
- Equilibrium verification is harder in this mode because the specimen responds simultaneously to three different loadings. Equation (2.19) must be checked *on each axis separately*.
- The constitutive interpretation requires a full 3D stress-path analysis. A uniaxial constitutive model cannot be fitted to data from this mode without producing spurious parameters.

Mode 6: Electromagnetic symmetric multidirectional impact

This is the most powerful and most physically distinctive mode of the Tri-HB facility. All six electromagnetic generators fire *simultaneously*, delivering identical pulses on the opposing bar pairs $(+X, -X)$, $(+Y, -Y)$, $(+Z, -Z)$. The two pulses on each axis arrive at the specimen at the same instant, superposing constructively in force while cancelling in momentum. The specimen is compressed from all sides simultaneously.

Mode 6: EM symmetric full XYZ



Sandstone, full XYZ symmetric, EM peak=400 MPa per bar, tau=200 us, 50 MPa hydrostatic prestress.
Peak specimen: 1184 MPa; peak bar: 400 MPa; mean rate: 251 /s

Figure 8.1: Mode 6 sample run: sandstone, full symmetric XYZ electromagnetic operation, $\sigma_{\text{peak}} = 400$ MPa per single bar and $\tau = 200 \mu\text{s}$.

Why this mode is special

- Force doubling at each face via constructive superposition.
- Zero net momentum transfer — the specimen does not accelerate.
- No transmission bar exists on the active axes; both $\pm$ -direction bars are incident-only.
- Bar stress is unchanged from uniaxial mode at the same σ_{peak} . Doubling occurs in the *specimen* stress, not the bar stress, which is a profound advantage for high-stress testing.

- Pure hydrostatic loading is possible when all six bars fire equally: the deviatoric stress $q \rightarrow 0$ and the specimen probes its volumetric (cap) response.

8.1 Symmetry of the wave field

Consider a single axis, say X. In symmetric mode both the $+X$ and $-X$ bars carry identical incident pulses arriving at the specimen face at the same instant. Define

$$\varepsilon_I^+(t) = \text{incident wave on the } +X \text{ bar, propagating } -x, \quad (8.1)$$

$$\varepsilon_I^-(t) = \text{incident wave on the } -X \text{ bar, propagating } +x. \quad (8.2)$$

By construction, in symmetric mode $\varepsilon_I^+ = \varepsilon_I^- = \varepsilon_I$ at the specimen position. After interaction with the specimen, each bar carries a reflected wave ε_R^+ or ε_R^- travelling back outward. The digital twin records both reflected waves explicitly so that the symmetric energy folding (2.35) can be applied directly.

8.2 Derivation of the symmetric-mode wave equations

The analogue of the three-wave equations (2.15)–(2.17) is derived in four steps.

Step A: Force at the specimen face from each bar

Each bar end exerts a force on the adjoining specimen face, given by the boundary-condition relation (2.11):

$$F^+(t) = E_b A_b [\varepsilon_I^+(t) + \varepsilon_R^+(t)], \quad (8.3)$$

$$F^-(t) = E_b A_b [\varepsilon_I^-(t) + \varepsilon_R^-(t)]. \quad (8.4)$$

Each force acts inward on the specimen.

Step B: Force balance on the specimen

Dynamic equilibrium of the specimen requires $F^+ = F^-$, otherwise the specimen would translate as a rigid body. By the symmetry of the input pulses this condition is automatic. The specimen axial stress is the force divided by the specimen cross-section:

$$\sigma_s^{\text{sym}}(t) = \frac{F^+ + F^-}{2A_s} = \frac{E_b A_b}{A_s} [\varepsilon_I(t) + \varepsilon_R(t)], \quad (8.5)$$

where the last equality uses $\varepsilon_I^+ = \varepsilon_I^- = \varepsilon_I$ and $\varepsilon_R^+ = \varepsilon_R^- = \varepsilon_R$ under perfect symmetry. The prefactor is $E_b A_b / A_s$, *double* that of the three-wave result (2.15), because two bars now drive the specimen instead of one.

Step C: Velocity of each specimen face

From (2.12), the $+X$ bar end velocity is

$$v_{\text{bar}}^+ = C_0 (\varepsilon_I^+ - \varepsilon_R^+) = C_0 (\varepsilon_I - \varepsilon_R),$$

which is the inward velocity of the $+X$ specimen face. By symmetry the $-X$ face moves inward at the same rate. The total rate of compression of the specimen is the sum of the two inward velocities divided by the specimen length:

$$\boxed{\dot{\varepsilon}_s^{\text{sym}}(t) = \frac{v^+ + v^-}{L_s} = \frac{2C_0}{L_s} [\varepsilon_I(t) - \varepsilon_R(t)].} \quad (8.6)$$

The strain rate is doubled relative to one-sided modes at the same σ_{peak} . The factor of 2 in (8.6) is purely geometric: both faces move inward, not just one.

Numerical note on initial conditions

In numerical implementation, the simplified driver $\dot{\varepsilon}_s^{\text{sym}}(t) \approx (2C_0/L_s)\varepsilon_I^+(t)$ is used to avoid a spurious deadlock at $t = 0$, where the reflected wave momentarily equals the incident wave before any specimen stress has built up. The simplified form becomes exact once equilibrium is reached.

Step D: Closure for the reflected wave

The reflected wave ε_R is obtained by inverting (8.5):

$$\boxed{\varepsilon_R(t) = \frac{\sigma_s(t) A_s}{E_b A_b} - \varepsilon_I(t).} \quad (8.7)$$

This is the analogue of the one-wave reduction $\varepsilon_T = \sigma_s A_s / (E_b A_b)$ of the standard analysis (§2.3.1); the formula is structurally identical, with the transmitted wave ε_T replaced by the reflected wave ε_R of the symmetric configuration.

Remark 8.1. An earlier internal draft of this derivation contained an erroneous factor of 2 in the denominator of (8.7), arising from a double-counting of the force balance. Equation (8.7) as stated here is consistent with the stress definition (8.5) and has been validated against the numerical implementation in `tri_hb_integrated.py`.

8.3 Comparison: standard vs symmetric mode

Quantity	Modes 1–5	Mode 6 (symmetric)
Specimen stress	$\sigma = \frac{E_b A_b}{2A_s} (\varepsilon_I + \varepsilon_R + \varepsilon_T)$	$\sigma = \frac{E_b A_b}{A_s} (\varepsilon_I + \varepsilon_R)$
Strain rate	$\dot{\varepsilon} = \frac{C_0}{L_s} (\varepsilon_I - \varepsilon_R - \varepsilon_T)$	$\dot{\varepsilon} = \frac{2C_0}{L_s} (\varepsilon_I - \varepsilon_R)$
Active bars	$\varepsilon_I, \varepsilon_R, \varepsilon_T$ (three)	$\varepsilon_I^\pm, \varepsilon_R^\pm$ (two pairs)
Effective drive	$\sigma_{\text{drive}} = \sigma_{\text{peak}}$	$\sigma_{\text{drive}} = 2\sigma_{\text{peak}}$
Max bar stress	$\sigma_{\text{bar}}^{\text{max}} \leq \sigma_{\text{peak}}$	$\sigma_{\text{bar}}^{\text{max}} \leq \sigma_{\text{peak}}$
Max specimen σ	$\sim \sigma_{\text{peak}}$	$\sim 2\sigma_{\text{peak}}$

The fundamental advantage of symmetric mode

The bar stress is the *same* in symmetric mode as in uniaxial mode at the same σ_{peak} , but the specimen experiences *double* the stress. Symmetric mode therefore reaches specimen stresses that would require bar yielding in any one-sided mode. For testing strong rocks under high confinement, this is the only way to access the high-stress regime without damaging the bars.

8.4 Symmetric sub-modes

The digital twin offers three configurations:

Symmetric $\pm X$ only

Two pulses on $\pm X$; Y and Z bars passive. The specimen experiences enhanced uniaxial loading — doubled stress and doubled strain rate — but no triaxial state. Useful as a baseline against which the multi-axis modes are compared.

Symmetric $\pm X \pm Y$ (biaxial)

Four pulses on $\pm X$ and $\pm Y$; Z passive. The specimen experiences a true biaxial dynamic squeeze. The deviatoric stress along Z drives any shear-related failure.

Full hydrostatic ($\pm X, \pm Y, \pm Z$)

All six pulses fire identically. Under perfect symmetry, $\sigma_x = \sigma_y = \sigma_z$ and the deviatoric stress $q = 0$; the specimen cannot fail by shear yielding. Instead it deforms volumetrically along its compaction cap. This probes the pressure–volume response captured by the curve $p(\varepsilon_v)$ with $p = \frac{1}{3}(\sigma_x + \sigma_y + \sigma_z)$ and $\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z$.

8.5 Pressure–volume response under hydrostatic loading

For full hydrostatic loading, the relevant constitutive curve is the compaction (cap) curve, governed by the bulk modulus K in the elastic regime and by an eventual cap pressure p_{cap} at which pore collapse and grain rearrangement take over [23]:

$$\frac{dp}{d\varepsilon_v} = K \quad (\text{initial elastic regime}), \quad p \rightarrow p_{\text{cap}} \quad (\text{post-cap plateau}). \quad (8.8)$$

In the current Streamlit constitutive surrogate, the cap pressure is rate-scaled using the same DIF convention as the axial strength model:

$$p_{\text{cap}} = 4\sigma_{c0} \text{DIF}, \quad \text{DIF} = 1 + b_{\text{rate}} \log_{10} \left(\frac{\max(\dot{\varepsilon}, \dot{\varepsilon}_{\text{ref}})}{\dot{\varepsilon}_{\text{ref}}} \right). \quad (8.9)$$

The digital twin automatically switches to this constitutive regime when the full-hydrostatic sub-mode is selected.

8.6 Procedure

Mode 6 procedure

1. Apply static pre-stress: set $\sigma_1^0 = \sigma_2^0 = \sigma_3^0$ for the hydrostatic sub-mode, or independent values otherwise.
2. Choose σ_{peak} (single-bar peak amplitude). Remember the specimen will experience $2\sigma_{\text{peak}}$ on each active axis. Verify $\sigma_{\text{peak}} < \sigma_{\text{prop}} = 1000 \text{ MPa}$.
3. Choose τ for the desired strain rate. Because the symmetric strain rate is doubled at the same τ , set $\tau \approx 2\tau_{\text{uniaxial}}$ for the same target $\dot{\varepsilon}$.
4. Charge the active capacitor banks to identical voltages.
5. Synchronise the triggers. Synchronisation jitter must satisfy $\Delta t_{\text{jitter}} \ll \tau/10$. Modern trigger electronics achieve $< 0.5 \mu\text{s}$.
6. Capture all bar gauge signals (four per active axis).
7. Verify symmetry: $\varepsilon_I^+ \approx \varepsilon_I^-$ and $\varepsilon_R^+ \approx \varepsilon_R^-$. Asymmetry indicates synchronisation failure or non-uniform specimen damage, and invalidates the doubled-stress analysis.
8. Apply (8.5) and (8.6) on each active axis.
9. For the hydrostatic sub-mode, compute $p(t)$, $\varepsilon_v(t)$, and plot $p(\varepsilon_v)$.
10. For the biaxial sub-mode, extract both the active-axis σ – ε curves and the passive-axis Poisson response.
11. Confirm bar plasticity: $\max_i |\sigma_{\text{bar},i}(t)| < \sigma_{\text{prop}}$.

8.7 Worked example: concrete under fully hydrostatic loading

A concrete specimen (UCS $\sigma_{c0} \approx 40 \text{ MPa}$) is to be tested under fully hydrostatic dynamic loading with target peak pressure $p_{\text{peak}} = 800 \text{ MPa}$. Static pre-stress $\sigma_1^0 = \sigma_2^0 = \sigma_3^0 = 50 \text{ MPa}$.

Required single-pulse amplitude. Since the specimen sees $2\sigma_{\text{peak}}$ on each axis, and the loading is hydrostatic so that the mean pressure equals the axial stress,

$$\begin{aligned} p_{\text{peak}} &= \sigma_1^0 + 2\sigma_{\text{peak}}, \\ 800 \text{ MPa} &= 50 \text{ MPa} + 2\sigma_{\text{peak}} \Rightarrow \sigma_{\text{peak}} = 375 \text{ MPa}. \end{aligned}$$

Bar stress check. Each bar carries one pulse at σ_{peak} :

$$\sigma_{\text{bar}}^{\text{max}} \approx 375 \text{ MPa} < 1000 \text{ MPa} \checkmark.$$

Plenty of margin: this configuration is safe even though the same specimen pressure would require $\sigma_{\text{peak}} \approx 750 \text{ MPa}$ in any one-sided mode, perilously close to the bar limit.

Pulse duration. For a target $\dot{\varepsilon} \sim 300 \text{ s}^{-1}$ in symmetric mode, the doubled strain-rate prefactor gives $\tau \approx 2C_0\varepsilon_I/(L_s \dot{\varepsilon}_{\text{target}}) \sim 200 \mu\text{s}$.

Result. Charge each of six capacitor banks to deliver $\sigma_{\text{peak}} = 375 \text{ MPa}$, $\tau = 200 \mu\text{s}$, fire synchronously with $\sigma_1^0 = \sigma_2^0 = \sigma_3^0 = 50 \text{ MPa}$ pre-applied. Expected outcome: pure compaction response, no shear failure, bar limits respected.

Chapter 9

Comparison of the six loading modes

9.1 Side-by-side comparison

Property	Mode 1 Gas-gun	Mode 2 Conf. cham- ber	Mode 3 Monash Tri- HB	Mode 4 EM uniax- ial	Mode 5 EM async	Mode 6 EM sym.
Pulse shape	Trap. (shaped)	Trap. (shaped)	Trap. (shaped)	Half-sine	Half-sine ×3	Half-sine ×6
Pulse source	Striker bar	Striker bar	Striker bar	EM coil	EM coil ×3	EM coil ×6
Specimen shape	Cylindrical / cubic	Cylindrical	Cubic	Cubic	Cubic	Cubic
Active dynamic bars	1 (X)	1 (X)	1 (X)	1 (X)	3 (+X, +Y, +Z)	Up to 6 (±XYZ)
Lateral architec- ture	None	Pressure chamber	4 output bars	4 output bars	4 output bars	4 output bars
Confining stress	None	$\sigma_2=\sigma_3=p_c$ axisym.	σ_2, σ_3 in- dep.	σ_2, σ_3 in- dep.	σ_2, σ_3 in- dep.	σ_2, σ_3 in- dep.
Default $\sigma_1, \sigma_2, \sigma_3$	0, 0, 0	30, 20, 20	30, 20, 15	30, 20, 15	30, 20, 15	30, 20, 15
Pre-stress range	0–50 MPa axial	0–50 MPa each	0–50 MPa each	0–50 MPa each	0–50 MPa each	0–50 MPa each
Pulse independence	$\sigma_{\text{peak}} \propto V$, $\tau \propto L$	$\sigma_{\text{peak}} \propto V$, $\tau \propto L$	$\sigma_{\text{peak}} \propto V$, $\tau \propto L$	Indep.	Indep.	Indep.
Specimen σ_{max}	$\sim \sigma_{\text{peak}}$	$\sim \sigma_{\text{peak}} + \sigma_1$	$\sim \sigma_{\text{peak}} + \sigma_1$	$\sim \sigma_{\text{peak}}$	$\sim \sigma_{\text{peak}}$ each axis	$\sim 2\sigma_{\text{peak}}$ each axis
Lode angle reach	$\theta = 0$	$\theta = 0$ fixed	$\theta \in [0^\circ, 60^\circ]$	$\theta = 0$	$\theta \in [0^\circ, 60^\circ]$	$\theta \in [0^\circ, 60^\circ]$
Stress-path control	Uniaxial only	Static triax. (axisym.)	Static triax. + dyn. axial	Uniaxial only	Yes (async)	Hydrostatic / biaxial
Pure hydrostatic?	No	No	No	No	Approx. if $\Delta t=0$	Yes
Best application	UCS, base- line SHPB	Standard confined SHPB	In-situ con- fined rock dyn.	Clean rate- dependence	Path- dependent failure	Cap me- chanics, high stress

Note on slider range. The 50 MPa pre-stress cap reflects the current digital-twin calibration. The slider maximum is set in one location (`max_conf` in `tri_hb_integrated.py`) and can be raised when the physical bench is upgraded.

9.2 Mode selection: matching the question to the apparatus

- **Mode 1 (Gas-gun, unconfined)** — when the research question is a baseline uniaxial dynamic strength, or a comparison against published classical SHPB data. Reliable and well-understood, but limited to a single loading axis with no active confinement.
- **Mode 2 (Confinement-chamber SHPB)** — when triaxial confinement is required but only axisymmetric (Lode angle $\theta = 0$) stress states are of interest. This is the most widely deployed triaxial-SHPB configuration in the dynamic-rock and dynamic-concrete literature,

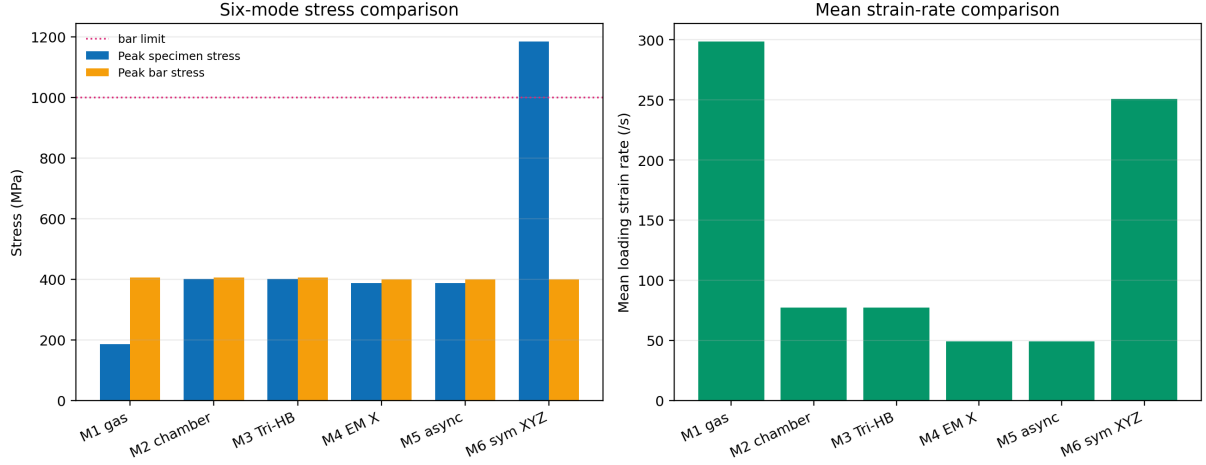


Figure 9.1: One-sample visual comparison of the six Tri-HB loading modes, using the same sandstone reference case to show how pulse source, confinement, stress path, and damage indicators change across the apparatus configurations.

well suited to high confining pressures and cylindrical specimens. Use this when the research plan does not call for independent σ_2 versus σ_3 .

- **Mode 3 (Monash Tri-HB)** — when in-situ stress conditions matter and the research requires independently controlled σ_2 and σ_3 , or a non-zero Lode angle: dynamic strength of rocks under realistic anisotropic triaxial confinement, blast-induced rockburst at depth, or any question where the full triaxial stress-path state is part of the physics [6, 7].
- **Mode 4 (EM Uniaxial)** — when rate-dependent constitutive behaviour is the primary question. The independent control of σ_{peak} and τ permits a controlled sweep of the $(\sigma, \dot{\epsilon})$ plane.
- **Mode 5 (EM Asynchronous)** — when loading-order effects are at issue. Particularly relevant for blast-induced rockburst and seismic loading. Pair with the Step 2 regime classifier and the Step 3 p - q - θ path.
- **Mode 6 (EM Symmetric)** — when (a) the target specimen stress exceeds the bar plasticity limit in any one-sided mode, or (b) the research targets the volumetric (cap) response of the material, which is inaccessible in any one-sided mode.

9.3 Common pitfalls and digital-twin diagnostics

- **Bar plasticity violation.** The three-tier warning system of §2.4 catches this in real time. Plan every physical test in the digital twin first.
- **Loss of stress equilibrium.** For very brittle specimens or very short pulses, the specimen may fail before reverberations equilibrate it. Check (2.19). Step 2 reports the equilibrium window $t_{\text{eq}} \in [3, 5] t_{\text{travel}}$ as a shaded band on the time-history plot.
- **Confusing strain rate with total strain.** A short, high-amplitude pulse delivers high $\dot{\epsilon}$ but possibly insufficient total strain to fail the specimen. Both metrics must be checked in the summary panel.
- **Mode-3 misinterpretation.** A uniaxial constitutive model cannot be fitted to data from Mode 5. Use a 3D stress-path-aware model (Drucker–Prager with cap, Mohr–Coulomb with

non-associated flow). The Step 3 p - q - θ diagnostics identify the path.

- **Symmetric-mode synchronisation failure.** If $\varepsilon_I^+ \neq \varepsilon_I^-$, the symmetry assumption is broken and the doubled-stress analysis is invalid.
- **Sign-convention mismatch.** The Step 1 reducer assumes the textbook convention ($\varepsilon_I > 0$, $\varepsilon_R < 0$, $\varepsilon_T > 0$ for a compression test). Use the compression-convention radio button to flip data acquired with the opposite polarity; an inverted convention typically appears as a negative absorbed energy, which the digital twin flags explicitly.

Chapter 10

The integrated computational workspace

The Virtual Tri-HB is delivered as a single integrated Streamlit application, `tri_hb_integrated.py`, that bundles four analysis steps under one shared experimental setup. This chapter documents the launch procedure, the navigation structure, and the export pipeline. The two chapters that follow document the equations of the experimental reducer (Chapter 11) and of the wave/damage/DEM module (Chapter 12).

10.1 Launching the application

Local launch

1. Install Python ≥ 3.9 .
2. Open a terminal in the folder containing `tri_hb_integrated.py`, `Tri-HB.py`, `wave_damage.py`, and `requirements.txt`.
3. Install dependencies: `pip install -r requirements.txt`.
4. Launch: `streamlit run tri_hb_integrated.py`.
5. Install Python ≥ 3.9 ; the current local test environment uses Python 3.13.
6. Open PowerShell or a terminal in the repository folder containing `tri_hb_integrated.py`, `Tri-HB.py`, `wave_damage.py`, and `requirements.txt`.
7. Install dependencies on Windows: `py -3.13 -m pip install -r requirements.txt`.
8. Launch the integrated workspace: `py -3.13 -m streamlit run tri_hb_integrated.py -server.port 8501`.
9. The browser opens at <http://localhost:8501>. If it does not, open the URL manually.

The integrated file `tri_hb_integrated.py` is the recommended entry point. It loads `Tri-HB.py` for the Step 1 configuration/simulator page and `wave_damage.py` for the Step 2–4 wave, stress-path, damage, energy, and validation pages. The direct command `streamlit run tri_hb_integrated.py` is equivalent when the `streamlit` executable is already on the system path. On macOS or Linux, replace `py -3.13` with the local Python launcher, commonly `python` or `python3`.

For shared use across a class, the repository can be pushed to GitHub and connected via `share.streamlit.io`. Students then access the integrated workspace through a public URL with no local installation.

10.2 Top-level navigation

A single radio in the sidebar selects which page is active. The intent is to move through them in order, but jumping back is supported because each step reads the shared `tri_hb_latest_config/tri_hb_latest_1` entries from the Streamlit session state.

Overview

Landing page that explains the workflow. Use it to confirm the handbook matches the deployed application version.

Step 1: Setup, simulator and data

Defines the shared experimental setup (material, specimen, bar, loading) and produces either a simulated stress-strain history or a reduced stress-strain history from uploaded gauge data. A second radio inside the sidebar switches between the simulator (Chapters 3–8) and the data reducer (Chapter 11).

Step 2: Wave model

Pulse timing, travel time, equilibrium window, and the wave-superposition / damage-memory regime classifier. Loads the `wave_damage.py` module with the “Wave model” tab first.

Step 3: Stress path and analysis

p - q - θ paths, invariant histories, comparison against the reduced experimental data from Step 1. Same module, “Stress path” tab first.

Step 4: Damage model and DEM validation

Damage evolution, stiffness and wave-speed degradation, energy balance, DEM/experimental descriptors, parametric study, and export. Same module, “Damage evolution” tab first.

10.3 Step 1 sidebar: the two tabs

Within the Step 1 simulator task, the sidebar is split into two tabs mirroring the operational separation a real test team uses.

Experimental setup

Specimen material (preset plus editable E , UCS, ν , ρ), specimen geometry (cross-section, side/diameter, loading length L_s), and bar properties (bar E_b , C_0 , cross-section). Defaults match the 42CrMo square 50 mm×50 mm bench; every value is editable.

Test loading

Loading mode (6 modes), symmetric-axis selector (Mode 6), pulse settings (striker velocity for the gas gun; σ_{peak} and τ for EM), static pre-stress sliders for $\sigma_1, \sigma_2, \sigma_3$, and the asynchronous timing sliders if Mode 5 is active.

Default pre-stress values (which the sliders populate when the loading mode is changed):

- Mode 1 (Gas-gun, unconfined): $\sigma_1 = \sigma_2 = \sigma_3 = 0$ MPa.
- Mode 2 (Confinement-chamber SHPB): $\sigma_1 = 30$ MPa, $\sigma_2 = \sigma_3 = p_c = 20$ MPa (axisymmetric, so the Y and Z sliders are linked to a single p_c control).
- Mode 3 (Monash Tri-HB): $\sigma_1 = 30$ MPa, $\sigma_2 = 20$ MPa, $\sigma_3 = 15$ MPa (three independent sliders).

- Modes 4–6 (electromagnetic): same independent defaults as Mode 3, $\sigma_1 = 30$ MPa, $\sigma_2 = 20$ MPa, $\sigma_3 = 15$ MPa.

When Step 1 is configured for “Experimental data analysis” instead of the simulator, the sidebar replaces the test-loading controls with the data-source selector and column-mapping controls described in Chapter 11.

10.4 Main panel tabs in the Step 1 simulator

The main panel updates live as inputs change. Five tabs are available; the fourth (p - ε_v) appears only in symmetric mode.

Bar waveforms

Time-history of all strain-gauge signals. The X-axis traces $\varepsilon_I, \varepsilon_R, \varepsilon_T$ are always shown when the loading mode places dynamic load on X. For Mode 3 (Monash Tri-HB), two additional passive lateral output-bar traces are plotted alongside the X-axis traces: ε_{out} (Y output bar) and ε_{out} (Z output bar), each representing the averaged signal of the two opposing output bars on that axis per (2.26)–(2.27). These lateral traces are the experimental record of the specimen’s Poisson-coupled lateral expansion, and serve as the diagnostic for damage-induced dilation. For Mode 2 (Confinement-chamber) the lateral output bars are absent and only the X-axis traces appear. For Mode 5 (asynchronous triaxial), three pairs of $\varepsilon_I, \varepsilon_R, \varepsilon_T$ are plotted, one pair per active axis. For Mode 6 (symmetric), both ε_I^+ and ε_I^- are plotted to verify synchronisation.

Specimen stresses

Total $\sigma_X, \sigma_Y, \sigma_Z$ on the specimen. In symmetric mode the mean pressure $p(t)$ is overlaid as a dashed reference. A horizontal dashed line at 1000 MPa indicates the bar proportional limit.

σ - ε curve

Loading branch only. The curve traces from the start of the pulse through peak strength and post-peak softening, stopping when strain plateaus.

p - ε_v

Visible only in symmetric mode. The compaction curve shows the initial bulk-modulus slope followed by the cap plateau.

Equations

Reference card showing the wave-analysis equations used by the active mode, including the standard three-wave method and the symmetric-mode variant.

10.5 Bar-plasticity warning banner

Whenever the maximum bar stress approaches the proportional limit, a coloured warning banner appears at the top of the page:

- **Caution** (yellow) at $0.85 \sigma_{\text{prop}}$.
- **Warning** (orange) at σ_{prop} : wave analysis becomes nonlinear; results unreliable.

- **Critical** (red) at $\sigma_Y = 1080$ MPa: bars deform plastically; the configuration must not be run on the physical apparatus.

This is the digital twin's primary safety feature. The banner should be clear (or at most cautionary) before any configuration is considered for physical testing.

10.6 Data export from Step 1

Four export buttons sit at the bottom of the simulator page. Filenames are auto-generated from the configuration so that re-running with different parameters produces uniquely-named files for batch comparison.

Button	Format	Contents
All Signals	CSV	Time-series: all bar gauge strains ($\varepsilon_I^\pm, \varepsilon_R^\pm, \varepsilon_T, \varepsilon_Y, \varepsilon_Z$ as available), specimen stresses ($\sigma_X, \sigma_Y, \sigma_Z$), strains, pressure, volumetric strain, axial strain rate.
σ - ε curve	CSV	Loading branch only: axial strain, axial stress, mean pressure, volumetric strain.
Configuration	JSON	Full input configuration, summary metrics, bar/rock properties, warning state, ISO 8601 export timestamp. Provides full provenance.
Full workbook	XLSX	Multi-sheet Excel: Sheet 1 Signals, Sheet 2 Stress-Strain, Sheet 3 Configuration.

Recommended export workflow

For coursework and research projects:

1. Run the simulator with the chosen configuration.
2. Verify the bar-plasticity banner is clear.
3. Export the JSON *and* either the CSV or XLSX, both under the auto-generated filename.
4. Move to Steps 2–4 to interpret wave timing, stress path, and damage; these inherit the Step 1 setup through session state.
5. Export Step 4's CSV for damage and energy histories alongside the Step 1 file. The JSON provides the audit trail linking both.

10.7 Worked Streamlit session

Session: granite at high confinement, EM symmetric XYZ

1. Top sidebar: select *Step 1: Setup, simulator and data*; keep the inner radio on *Test design and simulator*.
2. *Experimental setup* tab: *Material preset* = *granite*, keep auto-filled E , UCS, ν . Specimen cross-section *Square/cube*, side 50 mm, $L_s = 50$ mm. Leave bar settings at default.

3. *Test loading* tab: *Loading mode = EM Symmetric Multi-axis*; pick *Full hydrostatic* ($\pm X \pm Y \pm Z$).
4. $\sigma_{\text{peak}} = 400 \text{ MPa}$, $\tau = 200 \text{ }\mu\text{s}$, $\sigma_1 = \sigma_2 = \sigma_3 = 50 \text{ MPa}$.
5. Inspect the warning banner. If it shows “Caution” or above, reduce σ_{peak} until the banner is clear.
6. Open the $p\text{--}\varepsilon_v$ tab. Read the slope at low ε_v (dynamic bulk modulus K_d) and the plateau at high ε_v (cap pressure).
7. Click *Full workbook (XLSX)*; verify the *Configuration* sheet records all inputs.
8. Step 2: confirm the equilibrium band; check Δt^* (should be 0 for a synchronous test).
9. Step 3: verify the $p\text{--}q$ path collapses onto the $q \approx 0$ axis (hydrostatic).
10. Step 4: read the final damage D , central damage fraction D_c , and energy descriptors. Export the CSV for downstream comparison with DEM or experimental data.

Experimental data analysis and reduction

When the inner Step 1 radio is set to “Experimental data analysis”, the simulator’s loading controls are replaced by a data reducer that converts raw bar-gauge or direct stress–strain data into the same set of outputs the simulator produces. Students can therefore compare a real measurement and a simulation on the same axes, using the same downstream pages (Steps 2–4).

11.1 Three data sources

Latest simulator result

Reuses the most recent simulator output stored in session state. Useful for verifying agreement between the reducer and the simulator, and for feeding simulated data into the energy balance, wave model, and damage page.

Upload bar strain gauges

A CSV/XLSX with columns for time and the incident, reflected, and transmitted bar strains. The student maps each column manually and picks the strain unit (microstrain or strain), the time unit (microseconds, ms, or s), and the compression convention (positive or negative). The reducer then applies the SHPB equations and produces stress, strain, strain rate, and energy histories.

Upload direct stress–strain

A CSV/XLSX already containing a reduced stress and strain. Only unit conversion and a numerical derivative for the strain rate are needed.

11.2 Equations used by the reducer

Let E_b be the bar Young’s modulus, $C_0 = \sqrt{E_b/\rho_b}$ the bar wave speed, A_b the bar cross-section, A_s and L_s the specimen area and length, and $\varepsilon_I, \varepsilon_R, \varepsilon_T$ the incident, reflected, and transmitted bar strains.

Specimen stress: three-wave average

$$\sigma_s(t) = \frac{E_b A_b}{2A_s} [\varepsilon_I(t) + \varepsilon_R(t) + \varepsilon_T(t)]. \quad (11.1)$$

Specimen stress: transmitted-wave only

Assumes force equilibrium $\varepsilon_I + \varepsilon_R \approx \varepsilon_T$:

$$\sigma_s(t) = \frac{E_b A_b}{A_s} \varepsilon_T(t). \quad (11.2)$$

Specimen strain rate and strain

$$\dot{\varepsilon}_s(t) = \frac{C_0}{L_s} [\varepsilon_I(t) - \varepsilon_R(t) - \varepsilon_T(t)], \quad (11.3)$$

$$\varepsilon_s(t) = \int_0^t \dot{\varepsilon}_s(\tau) d\tau \quad (\text{trapezoidal}). \quad (11.4)$$

Under perfect equilibrium the rate reduces to $\dot{\varepsilon}_s = -2C_0\varepsilon_R/L_s$; the reducer retains the general three-wave form so that pre-equilibrium signals are still handled correctly.

Bar-wave energies

$$W_I(t) = E_b A_b C_0 \int_0^t \varepsilon_I^2 d\tau, \quad (11.5)$$

$$W_R(t) = E_b A_b C_0 \int_0^t \varepsilon_R^2 d\tau, \quad (11.6)$$

$$W_T(t) = E_b A_b C_0 \int_0^t \varepsilon_T^2 d\tau, \quad (11.7)$$

$$W_{\text{abs}}(t) = W_I(t) - W_R(t) - W_T(t). \quad (11.8)$$

The dimensional check is $A_b [\text{m}^2] \cdot E_b [\text{Pa}] \cdot C_0 [\text{m/s}] \cdot \int \varepsilon^2 dt [\text{s}] = \text{J}$.

Symmetric-mode (Mode 6) energy folding

Opposing bar pairs contribute incident *and* reflected pulses explicitly:

$$W_I = E_b A_b C_0 \int (\varepsilon_{I,+}^2 + \varepsilon_{I,-}^2) dt, \quad W_R = E_b A_b C_0 \int (\varepsilon_{R,+}^2 + \varepsilon_{R,-}^2) dt. \quad (11.9)$$

Absorbed energy from simulator stress–strain

$$W_{\text{abs}}(t) = \int_0^t \sigma_s(\tau) \dot{\varepsilon}_s(\tau) V_s d\tau, \quad V_s = A_s L_s. \quad (11.10)$$

Because the simulator stores σ_s as the *total* axial stress (static pre-load plus dynamic), the reflected, transmitted, and absorbed energies for a simulator-derived history are computed against the available incident energy. This monotone closure prevents the static-preload contribution from appearing as spurious wave energy and keeps the cumulative budget consistent with the incident envelope.

Direct stress–strain branch

Only unit conversion and a derivative are required:

$$\dot{\varepsilon}_s(t) = \frac{d\varepsilon_s}{dt} \quad (\text{central difference}). \quad (11.11)$$

Strain-unit options: strain (1), percent ($\div 100$), microstrain ($\times 10^{-6}$). Stress-unit options: MPa (pass-through), Pa ($\div 10^6$). The compression-convention selector applies a single global sign so that the formulas above remain consistent with the textbook convention $\varepsilon_I > 0$, $\varepsilon_R < 0$, $\varepsilon_T > 0$ for a compression test.

11.3 Reducer outputs and energy sanity check

The reducer writes its output to session state as `tri_hb_reduced_data` so that Steps 2–4 can use it for validation overlays. Outputs include time, stress, strain, strain rate, and (for bar-gauge data) the four energy histories.

A built-in energy sanity check raises a warning whenever any of

$$W_T > W_I, \quad W_R > W_I, \quad W_{\text{abs}} = W_I - W_R - W_T < 0 \quad (11.12)$$

holds. The most common cause is a sign or unit mismatch in the uploaded file (microstrain reported as strain; mixed compression-positive and compression-negative columns). The warning reports the numerical excess so that the student can judge how serious the mismatch is.

11.4 Worked reducer session

Session: comparing a measurement against the simulator

1. In Step 1, configure and run the simulator to match the geometry and loading of the physical test. Note the simulator's peak stress and strain rate.
2. Switch the inner Step 1 radio to "Experimental data analysis".
3. Sidebar: *Data source = Upload bar strain gauges*. The bar and specimen geometry inputs inherit the simulator setup automatically; override only if the upload was taken with a different rig.
4. Upload the CSV/XLSX. Map the time column and the three strain columns. Pick the time and strain units; check the compression convention against the raw data (textbook: $\varepsilon_I > 0$ for compression).
5. *Stress reduction = Three-wave average* for the baseline; then switch to *Transmitted-wave only* to confirm that the two formulas agree once equilibrium is reached.
6. Read the four metric cards: peak stress, peak strain, peak strain rate, absorbed energy. Inspect the time-history and energy tabs; if the energy warning appears, fix the sign or unit before continuing.
7. Move to Step 3. The "DEM/experimental descriptors" overlay (in Step 4) will use this reduced dataset for validation comparisons.

Chapter 12

Wave model, stress path, damage and DEM validation

Steps 2, 3, and 4 are three views into a single underlying module (`wave_damage.py`). They share the experimental setup defined in Step 1 — material, specimen length, pre-stress, pulse amplitude, pulse duration, loading dimensionality, asynchronous delay — and differ only in which analysis tab is opened first. The equations below are listed once and apply across all three steps.

12.1 Wave-model equations (Step 2 tab)

12.1.1 Elastic constants and wave speeds

Recalling (2.31)–(2.32), the constrained modulus and P-, S-wave speeds are

$$M = \frac{E(1 - \nu)}{(1 + \nu)(1 - 2\nu)}, \quad (12.1)$$

$$c_p = \sqrt{\frac{M}{\rho}}, \quad c_s = \sqrt{\frac{G}{\rho}}. \quad (12.2)$$

The wave travel time and the equilibrium window are

$$t_{\text{travel}} = \frac{L_s}{c_p}, \quad t_{\text{eq}} \in [3, 5] t_{\text{travel}}, \quad (12.3)$$

the latter shaded on the time-history plot.

12.1.2 Normalised delay and regime classifier

$$\Delta t^* = \frac{\Delta t}{t_{\text{travel}}}. \quad (12.4)$$

The digital twin partitions Δt^* into four physically distinct regimes:

Δt^*	Regime	Dominant physics
< 1	Synchronous wave superposition	Constructive overlap at specimen
1 to 3	Reverberation-coupled	Internal echoes still strong
3 to 10	Transitional	Decaying reverberations
> 10	Sequential, damage-memory	Each pulse acts on a damaged state

12.1.3 Superposition factor at the specimen centre

For a pulse envelope $g(\cdot)$ propagating from each end of the specimen, with right-bar amplitude ratio r ,

$$\sigma_{\text{centre}}(t) = \sigma_{x0} + A_x g\left(t - \frac{L_s}{2c_p}\right) + A_x r g\left(t - \Delta t - \frac{L_s}{2c_p}\right), \quad (12.5)$$

$$\eta_{\text{sup}} = \frac{\max_t \sigma_{\text{centre}}(t) - \sigma_{x0}}{A_x}. \quad (12.6)$$

A value $\eta_{\text{sup}} \rightarrow 2$ indicates full constructive overlap; $\eta_{\text{sup}} \rightarrow 1$ indicates sequential pulses.

Step 2: wave timing and interaction regime

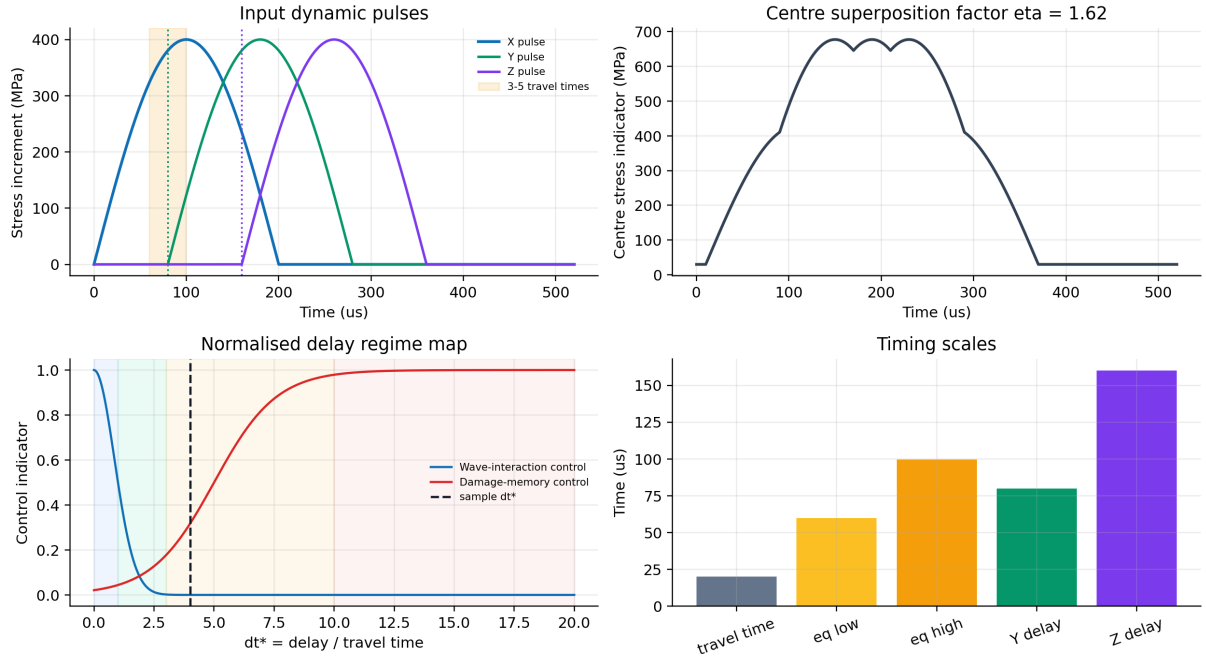


Figure 12.1: Step 2 sample wave model for the asynchronous XYZ case: input pulse timing, specimen-centre superposition, normalised delay regime map, and the travel-time/equilibrium scales used by the Streamlit workflow.

12.2 Stress-path equations (Step 3 tab)

Assuming the principal axes align with the bar axes, the diagonal Cauchy tensor has invariants

$$p = \frac{1}{3}(\sigma_x + \sigma_y + \sigma_z), \quad (12.7)$$

$$q = \sqrt{\frac{1}{2}[(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2]}, \quad (12.8)$$

$$J_2 = \frac{1}{2}(s_1^2 + s_2^2 + s_3^2), \quad s_i = \sigma_i - p, \quad (12.9)$$

$$J_3 = s_1 s_2 s_3, \quad (12.10)$$

$$\cos(3\theta) = \frac{3\sqrt{3}}{2} \frac{J_3}{J_2^{3/2}}. \quad (12.11)$$

The Lode angle θ distinguishes triaxial compression ($\theta = 0$) from triaxial extension ($\theta = 60^\circ$) and from pure shear ($\theta = 30^\circ$).

12.2.1 Failure surface and failure index

The failure surface is taken in the Hoek–Brown-like form

$$h(\theta) = 1 + a_\theta(1 - \cos 3\theta), \quad (12.12)$$

$$q_f(p, \theta) = (A + B p^n) h(\theta), \quad (12.13)$$

$$F(t) = \frac{q(t)}{q_f(t)}. \quad (12.14)$$

The failure index F is dimensionless; $F > 1$ means the stress state lies outside the static failure surface and damage is allowed to grow.

12.2.2 Equivalent (deviatoric) strain rate

Using elastic strains $\varepsilon_i = \sigma_i/E$ as a planning-grade estimate,

$$\dot{\varepsilon}_{eq} = \sqrt{\frac{2}{3} \cdot \frac{1}{2} [(\dot{\varepsilon}_x - \dot{\varepsilon}_y)^2 + (\dot{\varepsilon}_y - \dot{\varepsilon}_z)^2 + (\dot{\varepsilon}_z - \dot{\varepsilon}_x)^2]}. \quad (12.15)$$

This indicator ignores plastic strain and lateral coupling and should be used as a regime tag, not as a constitutive strain rate.

Step 3: stress path and failure index

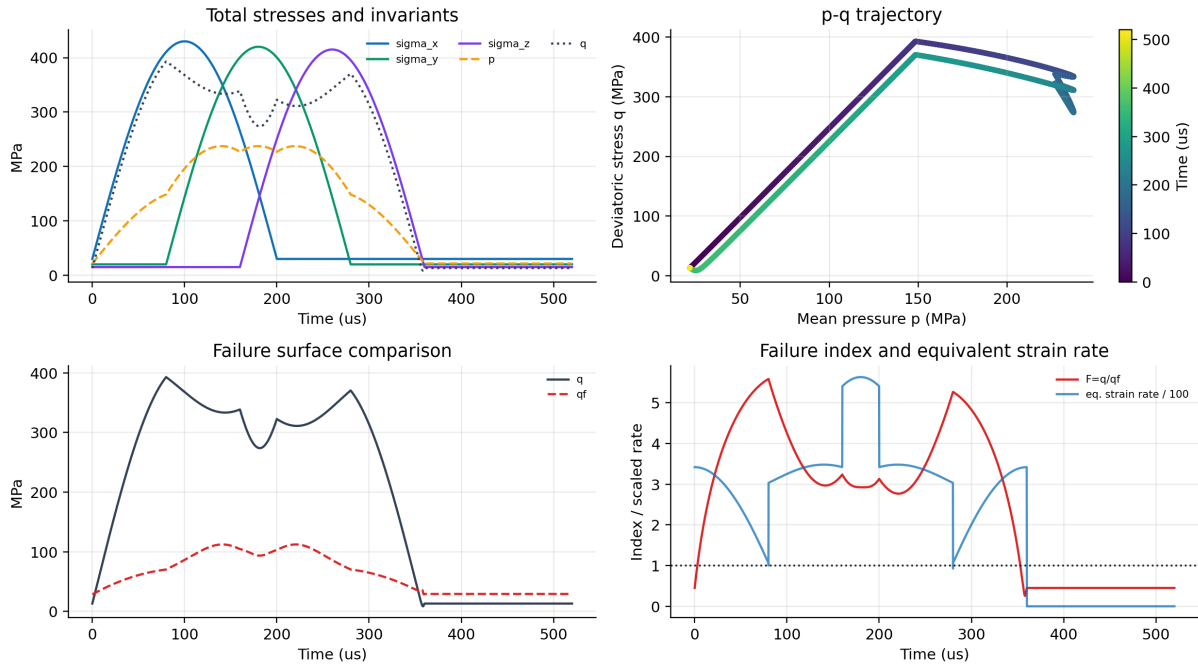


Figure 12.2: Step 3 sample stress-path analysis: total principal stresses, p - q trajectory, comparison of q with the pressure- and Lode-angle-dependent failure surface q_f , and the resulting failure index $F = q/q_f$.

12.3 Damage and energy equations (Step 4 tab)

12.3.1 Damage evolution

A Perzyna-type, saturating, rate-sensitive damage law is adopted [24, 25]:

$$\dot{D} = \frac{(1-D)^\alpha}{\tau_D} \left\langle \frac{F-1}{F_0} \right\rangle^m \left(\frac{|\dot{\epsilon}_{eq}|}{\dot{\epsilon}_0} \right)^\beta, \quad D \in [0, 1], \quad (12.16)$$

where $\langle \cdot \rangle$ denotes the Macaulay bracket (damage grows only when $F > 1$). The exponents and time scale are tabulated in Appendix A.

12.3.2 Stiffness and wave-speed degradation

$$E(D) = E_0(1-D), \quad c_p(D) = c_{p0} \sqrt{\max(1-D, 0)}. \quad (12.17)$$

Step 4: damage evolution and stiffness degradation

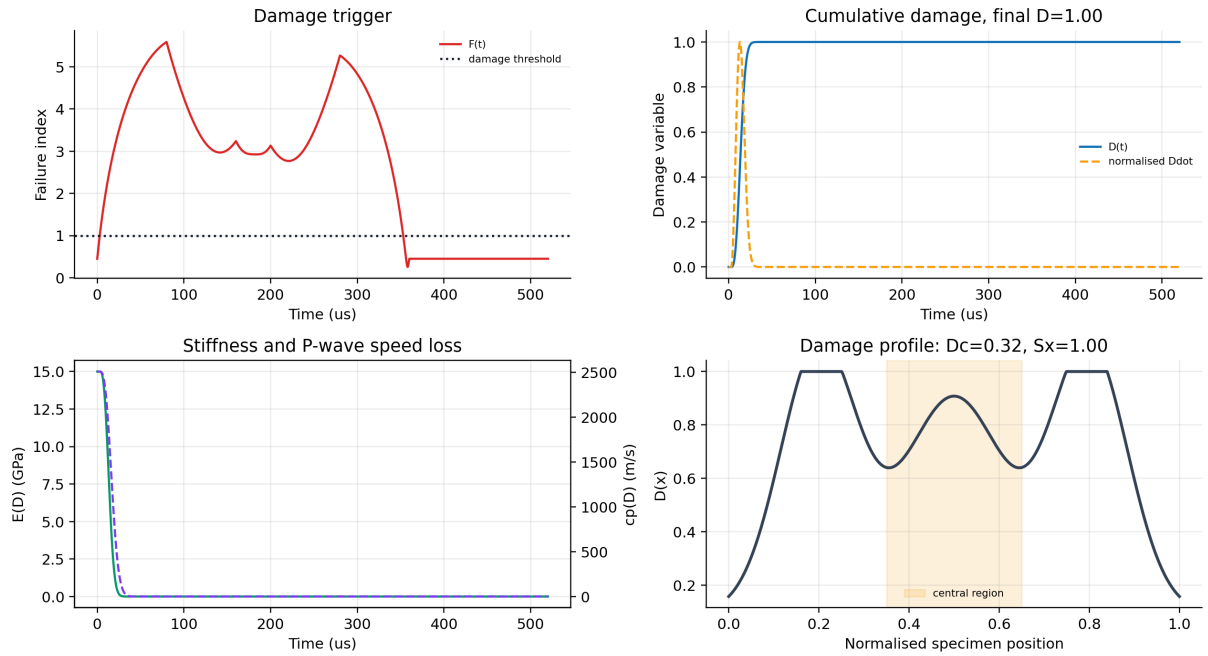


Figure 12.3: Step 4 sample damage analysis: failure-index trigger, cumulative damage and damage rate, stiffness and P-wave-speed degradation, and the synthetic damage-profile descriptors used for DEM/experimental comparison.

12.3.3 Energy indicators per unit volume

$$W_{el} = \frac{1}{2E} [\sigma_x^2 + \sigma_y^2 + \sigma_z^2 - 2\nu(\sigma_x\sigma_y + \sigma_y\sigma_z + \sigma_z\sigma_x)], \quad (12.18)$$

$$W_{input}(t) = \int_0^t (\sigma_x \dot{\epsilon}_x + \sigma_y \dot{\epsilon}_y + \sigma_z \dot{\epsilon}_z) d\tau, \quad (12.19)$$

$$W_{diss} \approx W_{input} - W_{el} + W_{el}(0). \quad (12.20)$$

With stresses in MPa and dimensionless strains, W is in MJ m^{-3} .

12.3.4 Synthetic DEM/experimental descriptors

The damage profile $D(x)$ across the specimen is reconstructed from three contributions (left wave, right wave, and central superposition), then the following summary statistics are derived:

$$D_c = \frac{\int_{\text{centre}} D(x) dx}{\int_0^{L_s} D(x) dx} \quad (\text{central damage fraction}), \quad (12.21)$$

$$S_x = 1 - \frac{|D_{\text{left}} - D_{\text{right}}|}{D_{\text{left}} + D_{\text{right}}} \quad (\text{symmetry index}). \quad (12.22)$$

These are model indicators intended for planning and interpretation, not substitutes for a real DEM simulation or CT scan; they should be validated against measured fracture patterns whenever such data are available.

Step 4: energy-density and validation indicators

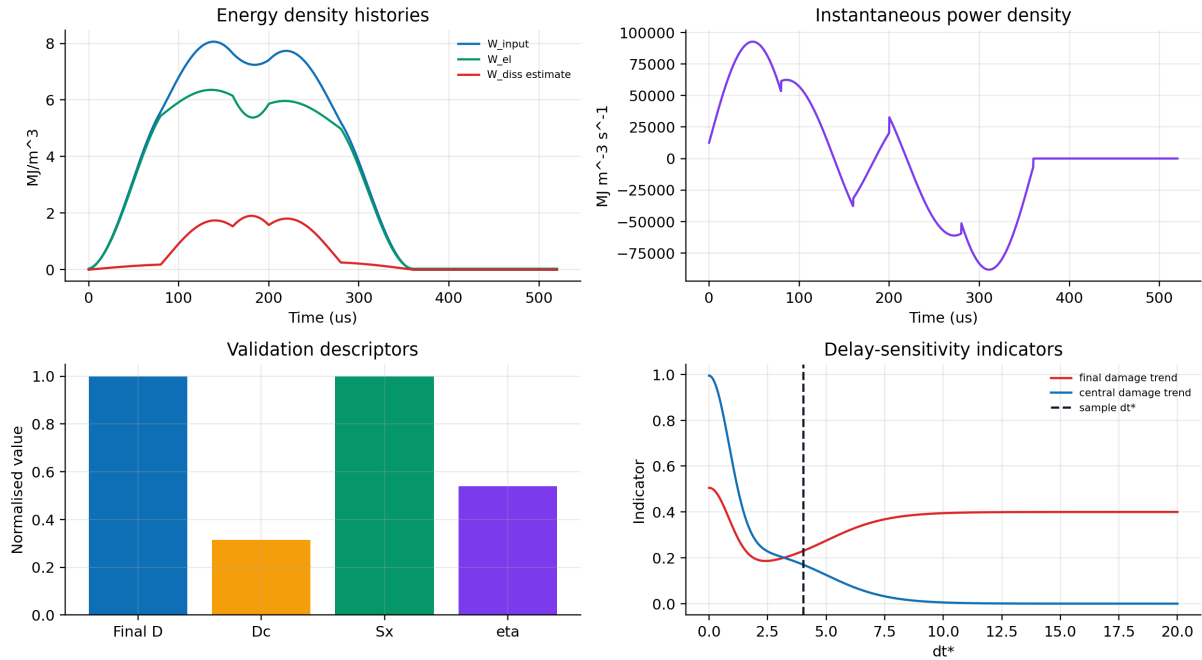


Figure 12.4: Step 4 sample energy-density and validation view: input, recoverable elastic, and dissipated energy-density estimates, instantaneous power density, scalar validation descriptors, and delay-sensitivity guide curves.

12.4 Sidebar controls common to Steps 2–4

Control level

Guided or Advanced. Advanced exposes the failure parameters A, B, n , the Lode amplitude a_θ , the damage time scale τ_D , the saturation exponent α , the overstress exponent m , the rate exponent β , the reference strain rate $\dot{\epsilon}_0$, and the normalisation F_0 .

Use Step 1 setup

When ticked (default if Step 1 has been run), the material, geometry, prestress, pulse amplitude, pulse duration, and loading dimensionality are read from the shared session state. Untick to enter a standalone setup.

Analysis goal

Quick validation, Stress-path focus, or Damage calibration. Pre-fills sensible defaults for the chosen workflow.

12.5 Export from Step 4

A single CSV download bundles every time-series produced by the wave/damage module: input pulses, $\sigma_x, \sigma_y, \sigma_z$, the invariants p, q, θ , the failure surface q_f , the failure index F , the equivalent strain rate, $D, \dot{D}, E(D), c_p(D), W_{\text{input}}, W_{\text{el}}, W_{\text{diss}}$.

Appendix A

Reference tables

A.1 Bar properties (42CrMo steel) – default values

These are the digital twin’s default bar parameters; each is editable in the Step 1 *Experimental setup* sidebar tab.

Quantity	Symbol	Default value
Young’s modulus	E_b	210 GPa
Density	ρ_b	7850 kg/m ³
Bar wave speed	C_0	5172 m/s
Proportional limit	σ_{prop}	1000 MPa
Yield stress (approx.)	σ_Y	1080 MPa
Acoustic impedance	$Z_b = \rho_b C_0$	$40.6 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
Round bar area (editable default $\Phi 50$)	A_b	1963 mm ²
Square bar area (50 mm×50 mm)	A_b	2500 mm ²

A.2 Default specimen parameters in the simulator

Material	E_s (GPa)	σ_{c0} (MPa)	ν	b_{rate}	$\dot{\epsilon}_{\text{ref}}$ (s ⁻¹)	k_{conf}	ϵ_{peak}
Sandstone	15	80	0.20	0.18	1×10^{-5}	4.5	0.008
Granite	50	180	0.25	0.12	1×10^{-5}	5.5	0.005
Concrete	30	40	0.20	0.22	1×10^{-5}	4.0	0.006

Selecting a preset in the Step 1 sidebar populates the editable E_s , σ_{c0} , and ν fields from this table. The remaining columns are hidden preset constants used by the Streamlit constitutive model.

A.3 Default damage-model parameters (Step 4 Advanced mode)

Symbol	Meaning	Default
A	Failure-surface intercept	15 MPa
B	Failure-surface pressure coefficient	1.3
n	Failure-surface pressure exponent	0.75
a_θ	Lode-angle amplitude	0.10
τ_D	Damage time scale	30 μ s
α	Saturation exponent	1.0
m	Overstress exponent	2.0
β	Rate exponent	0.15
$\dot{\epsilon}_0$	Reference strain rate	1.0 s ⁻¹
F_0	Failure-index normalisation	1.0

A.4 Useful identities

Acoustic impedance: $Z = \rho c$

Pulse transmission (Fresnel): $\frac{F_T}{F_I} = \frac{2Z_2}{Z_1 + Z_2}, \quad \frac{F_R}{F_I} = \frac{Z_2 - Z_1}{Z_1 + Z_2}$

Dynamic Increase Factor: $\text{DIF} = 1 + b_{\text{rate}} \log_{10}[\max(\dot{\epsilon}, \dot{\epsilon}_{\text{ref}})/\dot{\epsilon}_{\text{ref}}]$

Mean pressure: $p = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$

Deviatoric stress: $q = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}$

Lode angle: $\cos(3\theta) = \frac{3\sqrt{3}}{2} J_3 / J_2^{3/2}$

P-wave modulus: $M = E(1 - \nu)/[(1 + \nu)(1 - 2\nu)]$

Shear, bulk modulus: $G = E/[2(1 + \nu)], \quad K = E/[3(1 - 2\nu)]$

Appendix B

Suggested exercises

1. In Mode 1 (gas-gun), determine the maximum striker velocity for which a granite specimen with 50 MPa confinement remains within the bar proportional limit. Verify the hand calculation against the warning system of the digital twin.
2. In Mode 3 (gas-gun coupled triaxial), test sandstone ($\sigma_{c0} = 80$ MPa, $k_{\text{conf}} = 4.5$, $b_{\text{rate}} = 0.18$) at striker velocity $V = 20$ m/s and three confining pre-stress levels: $\sigma_2 = \sigma_3 = 0, 20$ MPa, and 40 MPa. Plot the predicted dynamic peak strength from (5.2) against confinement, and check whether the bar stress stays within σ_{prop} in each case. Discuss why confinement strengthens the rock dynamically without increasing the bar stress.
3. In Mode 4, keep $\sigma_{\text{peak}} = 400$ MPa fixed and vary τ from 50 μs to 300 μs . Plot peak specimen stress against strain rate. Extract the rate-hardening parameter b_{rate} and compare with the nominal value in Appendix A.
4. In Mode 5, run sandstone at $\sigma_{\text{conf}} = 30$ MPa with $\Delta t_y = \Delta t_z = 0$ (synchronous), then with $\Delta t_y = 100 \mu\text{s}$, $\Delta t_z = 200 \mu\text{s}$. Compare peak σ_x , then open Step 2 and record Δt^* and the regime classifier output. Discuss the link between Δt^* and the change in peak strength.
5. In Mode 6 (symmetric XYZ), find the peak specimen pressure achievable with single-pulse $\sigma_{\text{peak}} = 900$ MPa (right at the bar limit). Compare with the maximum pressure achievable in Mode 4 before bar yielding.
6. Derive equations (8.5) and (8.6) from first principles (force and velocity continuity), without consulting Chapter 8.
7. Extend the symmetric-mode derivation to the case where the two opposing pulses have different amplitudes, say σ_{peak}^+ and σ_{peak}^- . Show that the specimen experiences both a squeeze proportional to $\sigma_{\text{peak}}^+ + \sigma_{\text{peak}}^-$ and a net acceleration proportional to $\sigma_{\text{peak}}^+ - \sigma_{\text{peak}}^-$. At what amplitude ratio does this degenerate into Mode 4 (one-sided) behaviour?
8. In Mode 4, test sandstone with $\sigma_{\text{peak}} = 300$ MPa, $\tau = 200 \mu\text{s}$, $\sigma_2 = \sigma_3 = 30$ MPa. Run the simulation three times with $\sigma_1 = 0, 25$ MPa, and 50 MPa. Export the σ - ϵ CSV from each run, overlay the three curves, and verify that increasing σ_1 shifts the entire curve upward by a constant offset without changing peak strength. Explain why σ_1 behaves differently from σ_2 and σ_3 .
9. In Step 1, run any Mode 4 case. Switch the inner radio to “Experimental data analysis” and choose *Data source = Latest simulator result*. Confirm that the three-wave reduction reproduces the simulator’s stress-strain curve. Switch to *Transmitted-wave only* and verify that the two reductions agree once stress equilibrium is reached.
10. Take the same Mode 4 simulator result and download the All Signals CSV. Re-upload it through the “Upload bar strain gauges” branch with column mapping. Verify the reduced curve matches the simulator-derived curve, then deliberately mis-set the strain unit (e.g. claim “microstrain” for data stored as strain) and observe how the energy sanity warning fires.

11. Move to Step 3 with any asynchronous Mode 5 simulator result. Read Δt^* and the regime classifier off the Wave model tab, then trace the p - q path in the Stress path tab. Identify the instant at which $F(t) = q(t)/q_f(t)$ first exceeds unity and compare with the time at which damage starts to grow on the Damage evolution tab in Step 4.
12. Choose any test case of interest and download all four file types (Signals CSV, σ - ε CSV, Configuration JSON, Full XLSX). Open the JSON file in a text editor and identify: (a) the simulator version, (b) the export timestamp, (c) the bar plasticity warning state, (d) the peak specimen stress. Discuss how this metadata supports reproducibility in a research context.

References

- [1] H. Kolsky. “An investigation of the mechanical properties of materials at very high rates of loading”. In: *Proceedings of the Physical Society. Section B* 62.11 (1949), pp. 676–700. DOI: [10.1088/0370-1301/62/11/302](https://doi.org/10.1088/0370-1301/62/11/302).
- [2] U. S. Lindholm. “Some experiments with the split Hopkinson pressure bar”. In: *Journal of the Mechanics and Physics of Solids* 12.5 (1964), pp. 317–335. DOI: [10.1016/0022-5096\(64\)90028-6](https://doi.org/10.1016/0022-5096(64)90028-6).
- [3] P. S. Follansbee and C. Frantz. “Wave propagation in the split Hopkinson pressure bar”. In: *Journal of Engineering Materials and Technology* 105.1 (1983), pp. 61–66. DOI: [10.1115/1.3225620](https://doi.org/10.1115/1.3225620).
- [4] Q. B. Zhang and J. Zhao. “A review of dynamic experimental techniques and mechanical behaviour of rock materials”. In: *Rock Mechanics and Rock Engineering* 47.4 (2014), pp. 1411–1478. DOI: [10.1007/s00603-013-0463-y](https://doi.org/10.1007/s00603-013-0463-y).
- [5] H. C. Wang and Q. B. Zhang. “Experimental testing, characterisation methods, and mechanical behaviour of blast-induced rock damage from uniaxial strain to multiaxial stress states”. In: *International Journal of Rock Mechanics and Mining Sciences* 201 (2026), p. 106436. DOI: [10.1016/j.ijrmms.2026.106436](https://doi.org/10.1016/j.ijrmms.2026.106436).
- [6] K. Liu, Q. B. Zhang, G. Wu, J. C. Li, and J. Zhao. “Dynamic mechanical and fracture behaviour of sandstone under multiaxial loads using a triaxial Hopkinson bar”. In: *Rock Mechanics and Rock Engineering* 52.7 (2019), pp. 2175–2195. DOI: [10.1007/s00603-018-1691-y](https://doi.org/10.1007/s00603-018-1691-y).
- [7] Q. B. Zhang, C. H. Braithwaite, and J. Zhao. “Hugoniot equation of state of rock materials under shock compression”. In: *Philosophical Transactions of the Royal Society A* 375.2085 (2017), p. 20160169. DOI: [10.1098/rsta.2016.0169](https://doi.org/10.1098/rsta.2016.0169).
- [8] Heping Xie, Jianbo Zhu, Tao Zhou, and Jian Zhao. “Novel three-dimensional rock dynamic tests using the true triaxial electromagnetic Hopkinson bar system”. In: *Rock Mechanics and Rock Engineering* 54.4 (2021), pp. 2079–2086. DOI: [10.1007/978-3-030-60242-0_73](https://doi.org/10.1007/978-3-030-60242-0_73).
- [9] R. M. Davies. “A critical study of the Hopkinson pressure bar”. In: *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences* 240.821 (1948), pp. 375–457. DOI: [10.1098/rsta.1948.0001](https://doi.org/10.1098/rsta.1948.0001).
- [10] R. A. Graham. *Pochhammer–Chree dispersion in elastic cylindrical bars: practical correction notes*. Technical note (compiled summary). 1975.
- [11] D. J. Frew, M. J. Forrestal, and W. Chen. “Pulse shaping techniques for testing brittle materials with a split Hopkinson pressure bar”. In: *Experimental Mechanics* 42.1 (2002), pp. 93–106. DOI: [10.1007/BF02411056](https://doi.org/10.1007/BF02411056).
- [12] H. Kolsky. *Stress Waves in Solids*. New York: Dover Publications, 1963.
- [13] W. W. Chen and B. Song. *Split Hopkinson (Kolsky) Bar: Design, Testing and Applications*. New York: Springer, 2011. DOI: [10.1007/978-1-4419-7982-7](https://doi.org/10.1007/978-1-4419-7982-7).
- [14] G. T. Gray. “Classic split-Hopkinson pressure bar testing”. In: *ASM Handbook, Volume 8: Mechanical Testing and Evaluation*. ASM International, 2000, pp. 462–476.
- [15] K. F. Graff. *Wave Motion in Elastic Solids*. Oxford: Oxford University Press, 1975.

- [16] K. Ravi-Chandar and G. Subhash. “Dynamic testing of brittle materials – methodologies and interpretation”. In: *Mechanics of Materials* 35.3–6 (2003), pp. 257–275. DOI: [10.1016/S0167-6636\(02\)00280-2](https://doi.org/10.1016/S0167-6636(02)00280-2).
- [17] Y. X. Zhou, K. Xia, X. B. Li, H. B. Li, G. W. Ma, J. Zhao, Z. L. Zhou, and F. Dai. “Suggested methods for determining the dynamic strength parameters and mode-I fracture toughness of rock materials”. In: *International Journal of Rock Mechanics and Mining Sciences* 49 (2012). ISRM Suggested Method, first circulated 2010, pp. 105–112. DOI: [10.1016/j.ijrmms.2011.10.004](https://doi.org/10.1016/j.ijrmms.2011.10.004).
- [18] E. D. H. Davies and S. C. Hunter. “The dynamic compression testing of solids by the method of the split Hopkinson pressure bar”. In: *Journal of the Mechanics and Physics of Solids* 11.3 (1963), pp. 155–179. DOI: [10.1016/0022-5096\(63\)90050-4](https://doi.org/10.1016/0022-5096(63)90050-4).
- [19] Comité Euro-International du Béton. *CEB-FIP Model Code 1990*. Tech. rep. London: Thomas Telford, 1993.
- [20] D. L. Grote, S. W. Park, and M. Zhou. “Dynamic behavior of concrete at high strain rates and pressures: I. Experimental characterization”. In: *International Journal of Impact Engineering* 25.9 (2001), pp. 869–886. DOI: [10.1016/S0734-743X\(01\)00020-3](https://doi.org/10.1016/S0734-743X(01)00020-3).
- [21] Qian-Bing Zhang, Kai Liu, Gonglinan Wu, and Jian Zhao. “Dynamic Deformation, Damage, and Fracture in Geomaterials”. In: *Handbook of Damage Mechanics : Nano to Macro Scale for Materials and Structures*. Ed. by George Z. Voyiadjis. Cham: Springer International Publishing, 2022, pp. 379–422. ISBN: 978-3-030-60242-0. DOI: [10.1007/978-3-030-60242-0_73](https://doi.org/10.1007/978-3-030-60242-0_73).
- [22] X. B. Zhao, J. Zhao, J. G. Cai, and A. M. Hefny. “UDEC modelling on wave propagation across fractured rock masses”. In: *Computers and Geotechnics* 35.1 (2008). Cited in handbook for rate-dependent rock behaviour discussions, pp. 97–104. DOI: [10.1016/j.compgeo.2007.01.001](https://doi.org/10.1016/j.compgeo.2007.01.001).
- [23] A. N. Schofield. “Cambridge geotechnical centrifuge operations”. In: *Géotechnique* 30.3 (1980), pp. 227–268. DOI: [10.1680/geot.1980.30.3.227](https://doi.org/10.1680/geot.1980.30.3.227).
- [24] P. Perzyna. “Fundamental problems in viscoplasticity”. In: *Advances in Applied Mechanics* 9 (1966), pp. 243–377. DOI: [10.1016/S0065-2156\(08\)70009-7](https://doi.org/10.1016/S0065-2156(08)70009-7).
- [25] S. Murakami. *Continuum Damage Mechanics: A Continuum Mechanics Approach to the Analysis of Damage and Fracture*. Dordrecht: Springer, 2012. DOI: [10.1007/978-94-007-2666-6](https://doi.org/10.1007/978-94-007-2666-6).